K-FIELD DERIVED PROPERTIES

A Foundational Educational Reference on Cell3 Geometry, Anisotropic Carrier Structure, Weak Matter Coupling, Spin-Translation Response, Continuous-Rim Closure, and Earth Kinematic Calibration

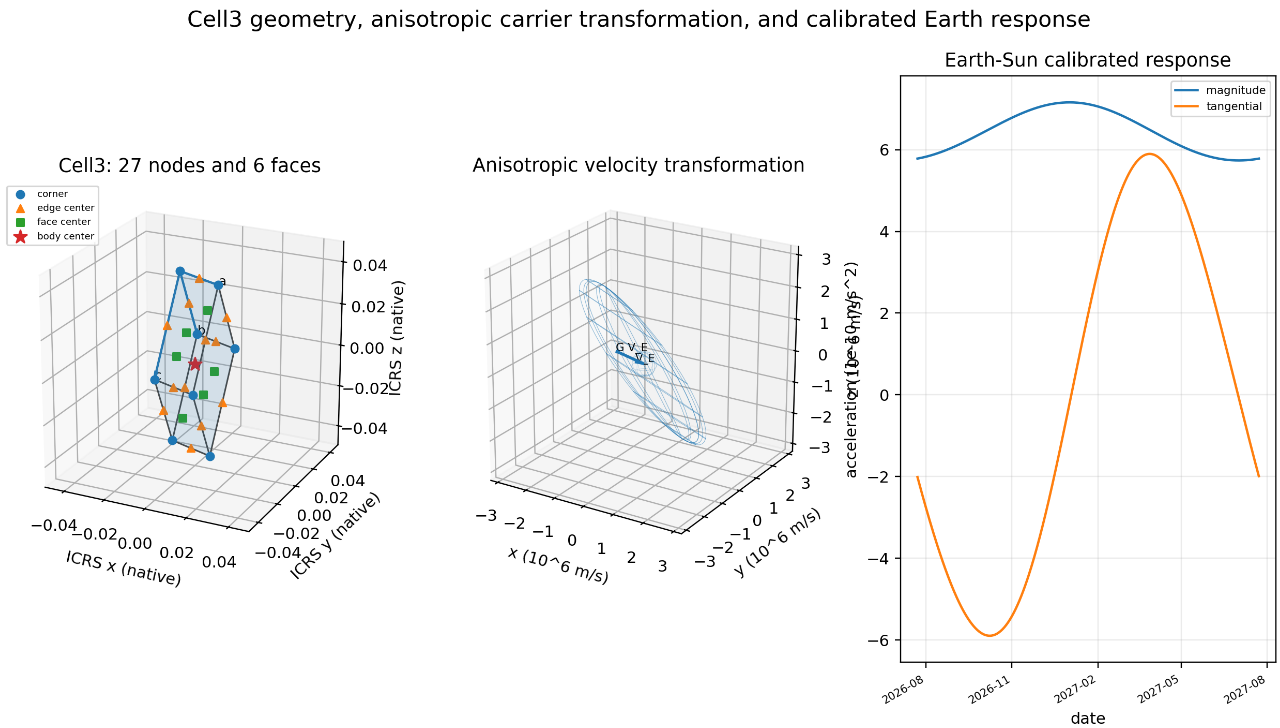

Figure 1. Title composite generated exclusively from the authoritative numerical source arrays.

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Archival foundational education reference - Unix epoch 1784893730 - 2026-07-24T11:48:50Z

Frontispiece: authoritative base cell, planes, and all nodes

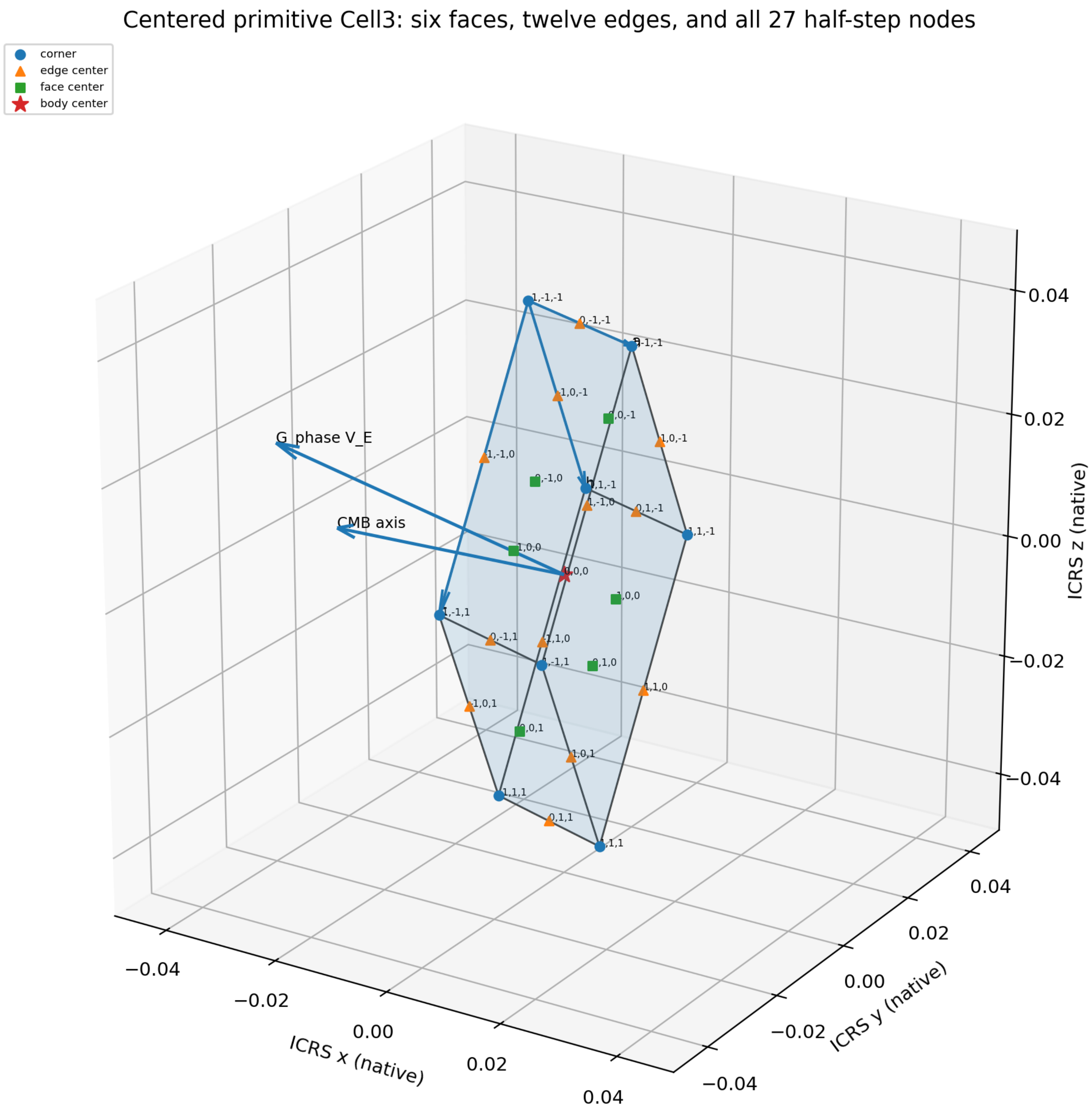

Figure 2. Dimensionally correct centered primitive Cell3 in ICRS native coordinates. All six boundary planes, twelve cell edges, twenty-seven half-step nodes, integer addresses, and directional references are present.

Abstract

This reference develops the derived properties of the K-Field framework as a self-contained hierarchy. The hierarchy begins with the exact 27-node Cell3 grammar, embeds that grammar in an oblique metric body, transforms a cosmological translation state through the symmetric positive-definite tensor G_phase, and couples the transformed state to rotating matter through a dimensionless coefficient kappa_F. The calibrated constitutive law is a_K = kappa_F H[omega cross (G_phase V_CMB)]. Its measurable response is therefore the product of four distinct objects: a material coupling, a body-geometry operator, an angular-velocity vector, and an anisotropically transformed translation vector.

The Earth-only kinematic calibration gives kappa_F=3.0740508997e-11, equivalent to 30.74051 parts per trillion. The tensor G_phase has eigenvalues 1.429328753, 2.112545026, and 8.800336435, with principal gain ratio 6.156972:1. It maps the declared Earth-CMB speed 369812.502 m/s to a transformed magnitude 930825.217 m/s. The cross product makes the leading response transverse, spin-direction-sensitive, and rank two: the response vanishes for spin parallel to G_phase V_CMB and is maximal for perpendicular spin.

The continuous-rim derivation provides an exact body-scale closure. Antipodal integration removes the radius-dependent rotational self-term and leaves F=(kappa_F M omega/2)n cross(G_phase V_E). The resultant is linear in total mass and angular rate, independent of radius at fixed angular rate for an ideal uniform zero-thickness rim, and torque-free about the rim center at the analyzed constitutive order. A one-kilogram rim at maximum orientation produces 14.307 micro-newtons per radian per second after calibration. The corresponding matter-field interface has high kinematic impedance.

The framework does not yet supply a field displacement variable, restoring-energy law, stress-energy tensor, propagation equation, spatial Green function, or reciprocal momentum equation. It therefore does not determine intrinsic elastic stiffness, energy density, propagation speed, interaction range, dispersion, or backreaction. The narrow supported conclusion is that the calibrated model describes a strongly structured carrier with a very weak matter-interface susceptibility. All quantitative statements in this paper are model-derived from the declared source data unless a public astronomical source is identified explicitly.

Central conclusion: kappa_F measures the susceptibility of matter to the locked K-Field kinematic tensor. It is not, by itself, an elastic modulus or a measurement of intrinsic field stiffness.

Contents

1. Reading the Framework as a Typed Scientific Hierarchy

The K-Field framework becomes understandable when each mathematical object is kept in its own role. A coordinate identifies a location. A metric defines lengths and angles. A reciprocal metric reads plane response and sensitivity. A tensor transforms direction-dependent motion. A cross product selects a transverse response direction. A material coefficient controls how strongly matter manifests that response. Mixing these roles produces incorrect physical interpretations, especially the mistaken identification of a small coupling coefficient with a weak carrier or a low elastic stiffness.

The hierarchy used in this reference is: exact finite address grammar -> metric Cell3 body -> reciprocal geometry -> anisotropic carrier tensor -> cosmic translation state -> spin-coupled kinematic kernel -> body-geometry filter -> material susceptibility -> observable acceleration or force. A downstream object can be calculated from upstream objects only through the declared transformation. This discipline is the basis for dimensional consistency and reproducibility.

Table 1. Typed object hierarchy.

Layer	Object	Meaning
Primitive grammar	L in {-1,0,1}^3	Exact 27-state centered address alphabet.
Metric embedding	p=(1/2)B L; M=B^T B	Converts addresses into native lengths, angles, areas, and volume.
Reciprocal embedding	R=M^-1	Reads inverse-distance-squared and plane-sensitivity structure.
Carrier transform	g=G_phase V_CMB	Maps cosmic translation through a fixed anisotropic positive-definite tensor.
Spin kernel	omega cross g	Produces a transverse, spin-odd kinematic acceleration scale.
Body geometry	H[omega cross g]	Filters the kernel through normalized mass geometry.
Matter interface	kappa_F	Dimensionless susceptibility multiplying the acceleration template.
Observable	a_K or F=M a_K	Acceleration or force predicted within the calibrated constitutive model.

1.1 Scientific status and scope

This paper is an internally complete statement of the declared model and its derived consequences. The Cell3 and K-Field quantities are not presented as established components of standard physics. The calibration is an attribution within the model: it asks what value of kappa_F makes the calculated K-indexed Earth response match selected fitted Earth kinematic channels. Independent experimental reproduction, controlled null tests, and a reciprocal field dynamics are required before the framework can be treated as an empirically established replacement for existing theories.

This distinction does not weaken the mathematical results. The geometric identities, tensor eigenanalysis, continuous-rim integral, dimensional scaling laws, and numerical reconstructions can be tested exactly from the supplied arrays. What remains open is whether nature follows the declared constitutive law and whether the Earth calibration isolates a new interaction rather than absorbing unmodeled conventional dynamics.

2. Mathematical Tools for the Technical Reader

2.1 Vectors, tensors, and units

A vector has magnitude and direction. In this report, V is a velocity vector in metres per second, omega is an angular-velocity vector in radians per second, and a is an acceleration vector in metres per second squared. Radians are dimensionless in SI dimensional analysis, so omega carries inverse seconds. The matrix G_phase is dimensionless. Therefore G_phase V has velocity units, and omega cross (G_phase V) has acceleration units.

```
V_CMB                 [m/s]
G_phase               [dimensionless]
g = G_phase V_CMB     [m/s]
omega cross g          [m/s^2]
H                      [dimensionless geometry operator]
kappa_F                [dimensionless material susceptibility]
a_K = kappa_F H[omega cross g]   [m/s^2]
```

A symmetric positive-definite matrix has three positive eigenvalues and three mutually orthogonal eigenvectors. Along each eigenvector, the matrix acts as a simple scalar gain. The eigenvalues therefore provide the cleanest summary of directional amplification. Positive definiteness does not make the force sign positive; the cross product and spin direction control the signed response.

2.2 Cross-product geometry

For any vectors x and y, x cross y is perpendicular to both. Its magnitude is |x||y|sin(theta), where theta is their included angle. Applied to the K-Field kernel, omega cross g vanishes when omega is parallel or antiparallel to g and reaches maximum magnitude when omega is perpendicular to g. Reversing spin changes omega to -omega and reverses the response exactly.

```
a_K dot omega = 0
a_K dot g = 0                  for H proportional to I
omega parallel g  =>  a_K = 0
omega perpendicular g  =>  |a_K| is maximal
omega -> -omega  =>  a_K -> -a_K
```

For a spherical body H=I/3, so H preserves directions and scales the vector by one third. For a general body, H can change component weights while preserving the distinction between the kinematic kernel and the material coefficient.

2.3 Static stiffness versus kinematic susceptibility

Static stiffness relates force to displacement, as in F=-k x. Viscous damping relates force to velocity, as in F=-c v. The leading K-Field law contains neither a field displacement nor a velocity-opposing drag term. It is bilinear in spin and a transformed background translation. The measured kappa_F is therefore a kinematic susceptibility: it tells how much of the declared acceleration template appears in matter.

A small kappa_F can coexist with a large or rigid carrier. The calibration constrains the matter-carrier interface, not a field modulus.

3. Cell3: The Geometric Foundation

Cell3 is an oblique primitive cell generated by three ordered basis vectors. The canonical local frame places a on the x axis, b in the x-y plane, and c in three dimensions. A common rigid rotation carries the same intrinsic cell into ICRS. Lengths, included angles, face areas, volume, direct metric, reciprocal metric, and the exact address grammar are unchanged by this rotation.

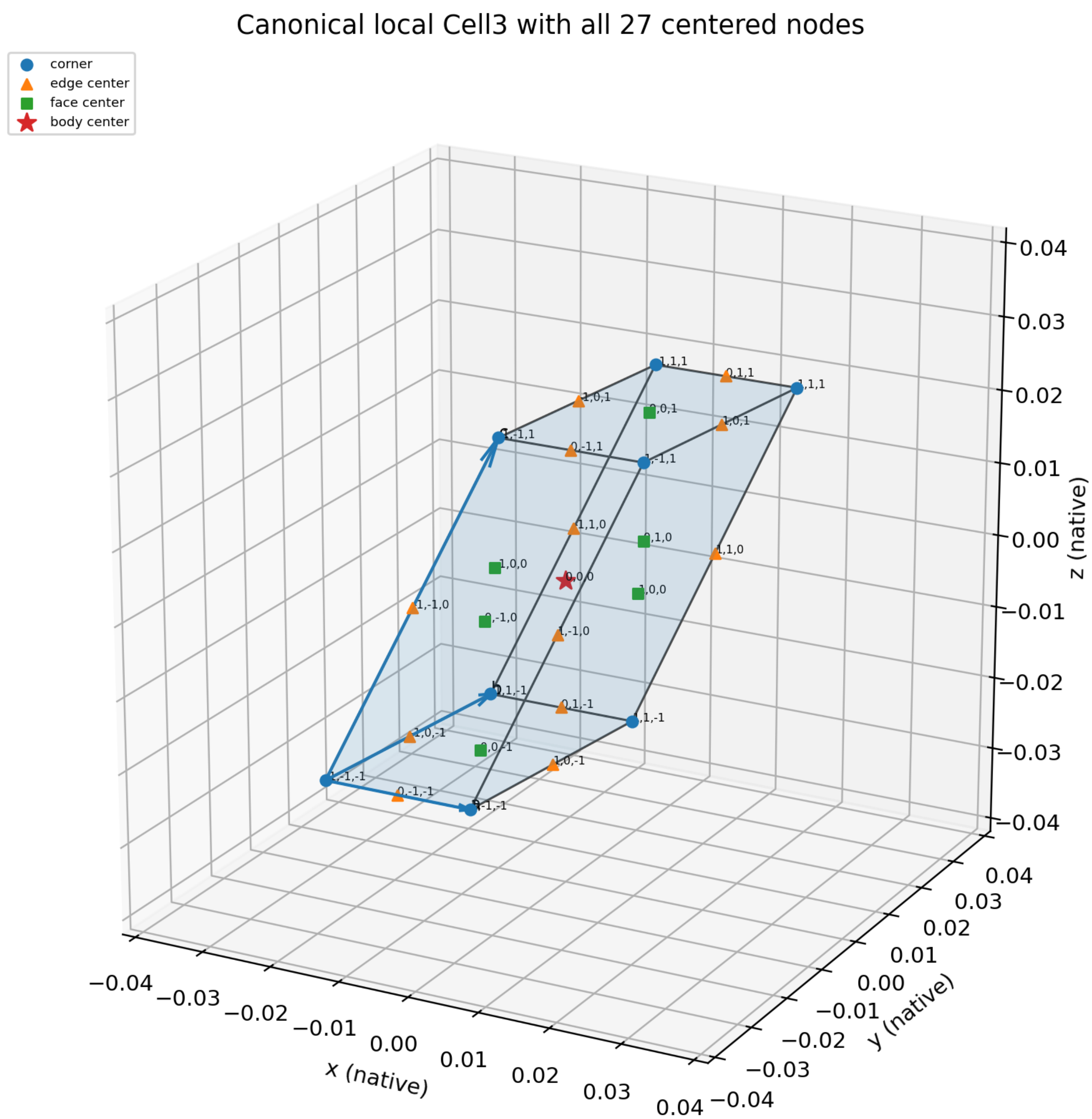

Figure 3. Canonical local Cell3 generated from the authoritative basis. All axes use native length units and all 27 centered nodes are labelled.

Table 2. Canonical Cell3 basis vectors.

Vector	x	y	z	length
a	0.02079705878161049	0	0	0.02079705878161049
b	0.003856816104884186	0.03628983297903845	0	0.03649420513058752
c	0.008182284351835403	0.03076238244505114	0.03765714700071637	0.04930856590029204

Table 3. Intrinsic Cell3 dimensions.

Quantity	Value	Units or meaning
|a|	0.02079705878161049	native length
|b|	0.03649420513058752	native length
|c|	0.04930856590029204	native length
theta_ab	83.933483012313 deg	included angle
theta_ac	80.448130018338 deg	included angle
theta_bc	50.363231813756 deg	included angle
A_ab	0.0007547217896398895	native^2
A_ac	0.001011255757656946	native^2
A_bc	0.001385784431740138	native^2
volume	2.842066937711308e-05	native^3

3.1 The exact 27-node grammar

The complete local node set uses three states along each primitive coordinate. In uncentered form, r=(u,v,w) with u,v,w in {0,1/2,1}. Doubling gives Q in {0,1,2}^3. Subtracting the body-center address gives L=Q-(1,1,1) in {-1,0,1}^3. The physical centered coordinate is p=(1/2)B L. The 27 nodes divide into eight corners, twelve edge centers, six face centers, and one body center.

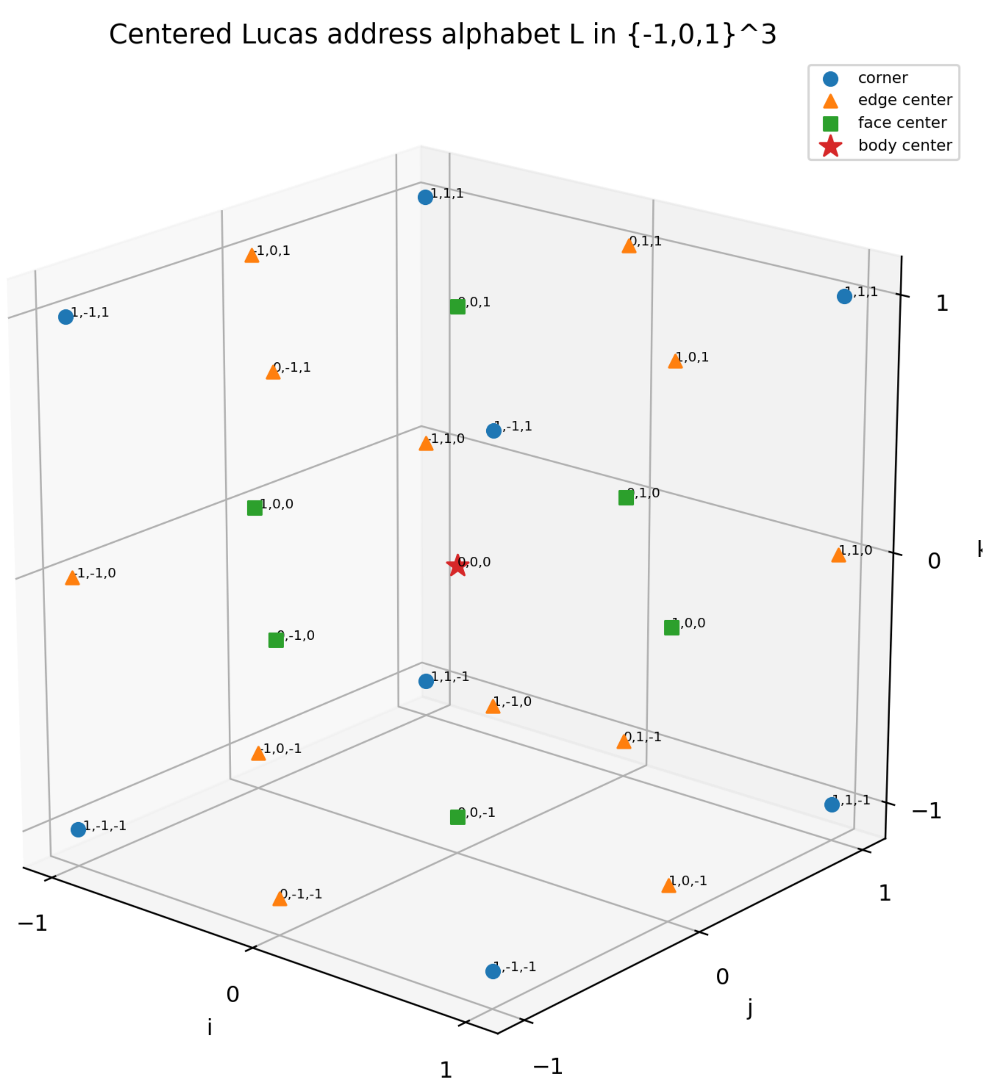

Figure 4. The centered integer address cube. The physical Cell3 is obtained by the linear metric embedding p=(1/2)B L.

Table 4. Node classes in centered address form.

Class	Address condition	Count
Corners	no zero entries in L	8
Edge centers	one zero entry in L	12
Face centers	two zero entries in L	6
Body center	L=(0,0,0)	1

The sheet index lambda=i+j+k partitions the node set into seven constant-sum sheets with counts 1, 3, 6, 7, 6, 3, 1. Under the declared polarity chi=(-1)^(i+j+k), every node in a given sheet has the same sign and adjacent sheets have opposite signs. This finite grammar is exact; floating-point coordinates enter only when B embeds it in the oblique metric.

3.2 Boundary planes and face normals

The six boundary faces form three opposite pairs. An ab face holds the c coefficient fixed at plus or minus one half; an ac face fixes b; a bc face fixes a. In ICRS, each plane is written n dot x + d = 0 using the source unit normal n and signed offset d. The offsets have native length units.

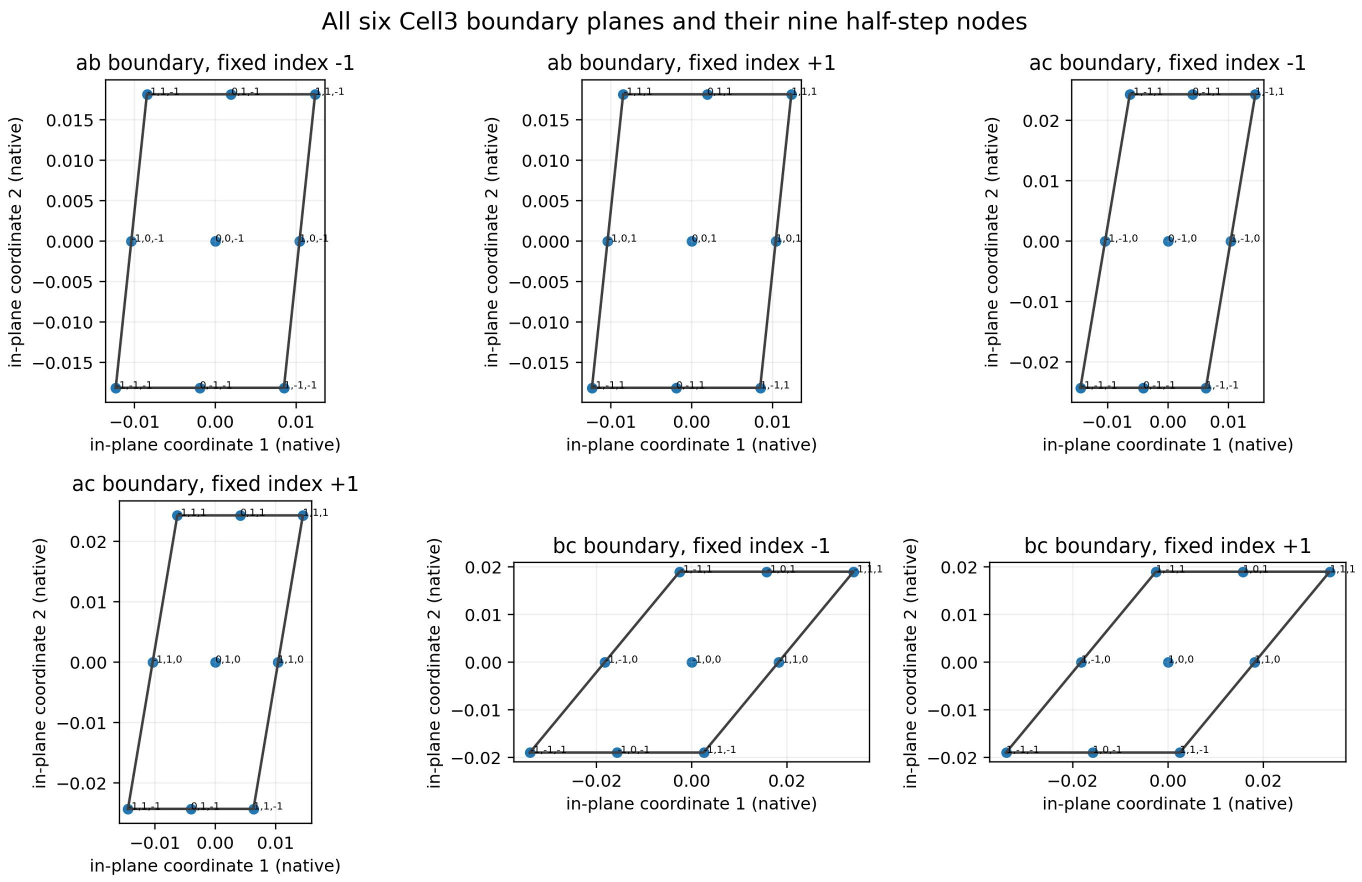

Figure 5. All six boundary planes shown in distance-preserving intrinsic coordinates. Each face contains nine labelled nodes.

Table 5. Complete six-plane equation ledger.

Plane family	Side	Unit normal n	d (native)	Equation
ab faces (constant c coefficient)	1	[0.1940922008, 0.9131771688, 0.3583792348]	-0.01882857350019	n dot x + d = 0
ab faces (constant c coefficient)	2	[0.1940922008, 0.9131771688, 0.3583792348]	0.01882857350019	n dot x + d = 0
ac faces (constant b coefficient)	1	[0.0890525941, 0.8665767693, -0.4910339483]	0.01405216690339	n dot x + d = 0

Plane family	Side	Unit normal n	d (native)	Equation
ac faces (constant b coefficient)	2	[0.0890525941, 0.8665767693, -0.4910339483]	-0.01405216690339	n dot x + d = 0
bc faces (constant a coefficient)	1	[-0.9959651903, 0.08360216251, -0.03261929056]	-0.01025436161905	n dot x + d = 0
bc faces (constant a coefficient)	2	[-0.9959651903, 0.08360216251, -0.03261929056]	0.01025436161905	n dot x + d = 0

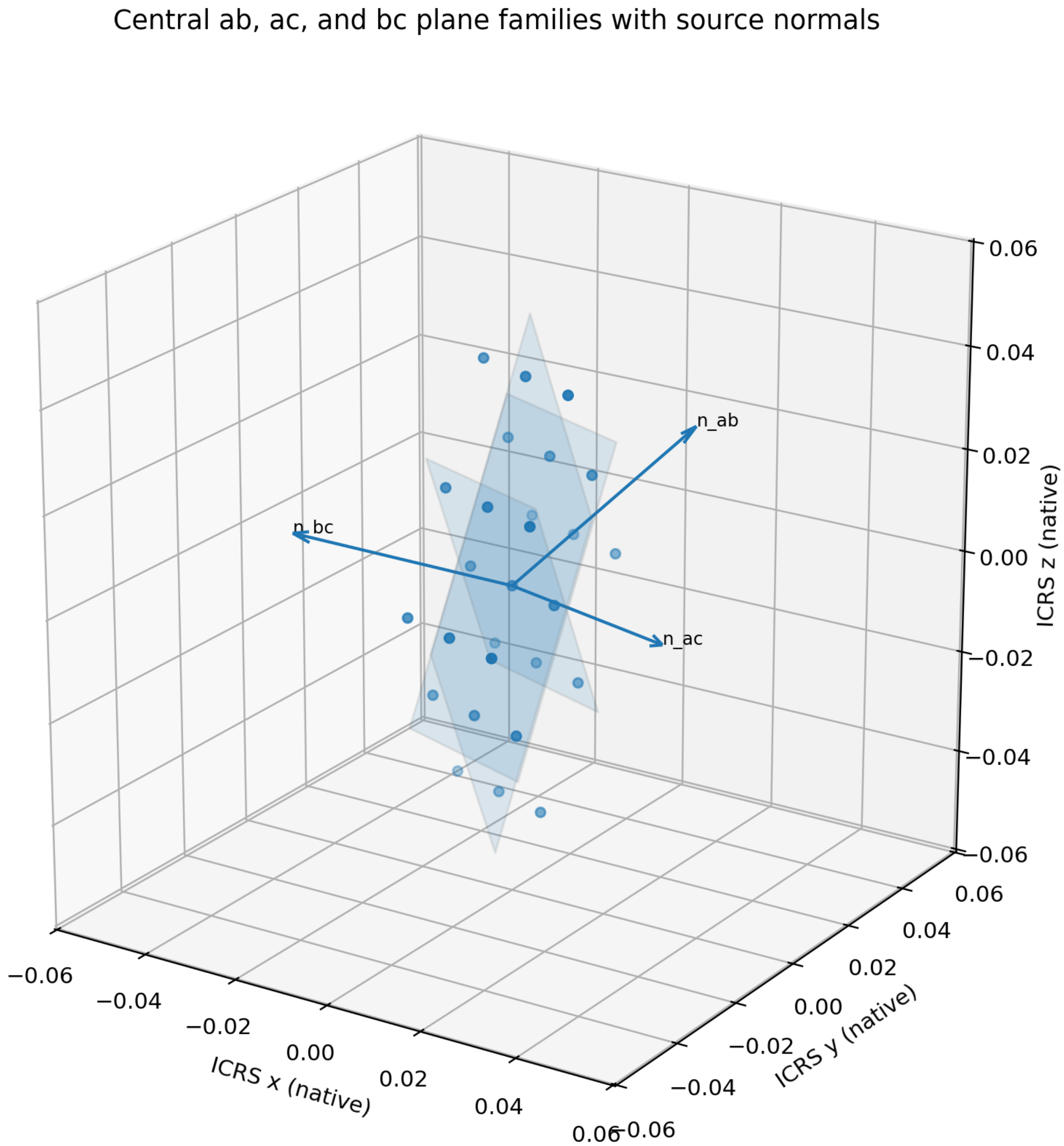

Figure 6. Central ab, ac, and bc plane families with all 27 ICRS nodes and the source face normals.

3.3 Direct metric, reciprocal metric, and the constant-trace pathway

The direct metric M=B^T B is the intrinsic directional ruler of Cell3. Its diagonal entries are squared edge lengths. Its off-diagonal entries are dot products and therefore encode angular shear. The reciprocal metric R=M^-1 is the inverse ruler used by reciprocal planes, q-values, and first-order volume sensitivity.

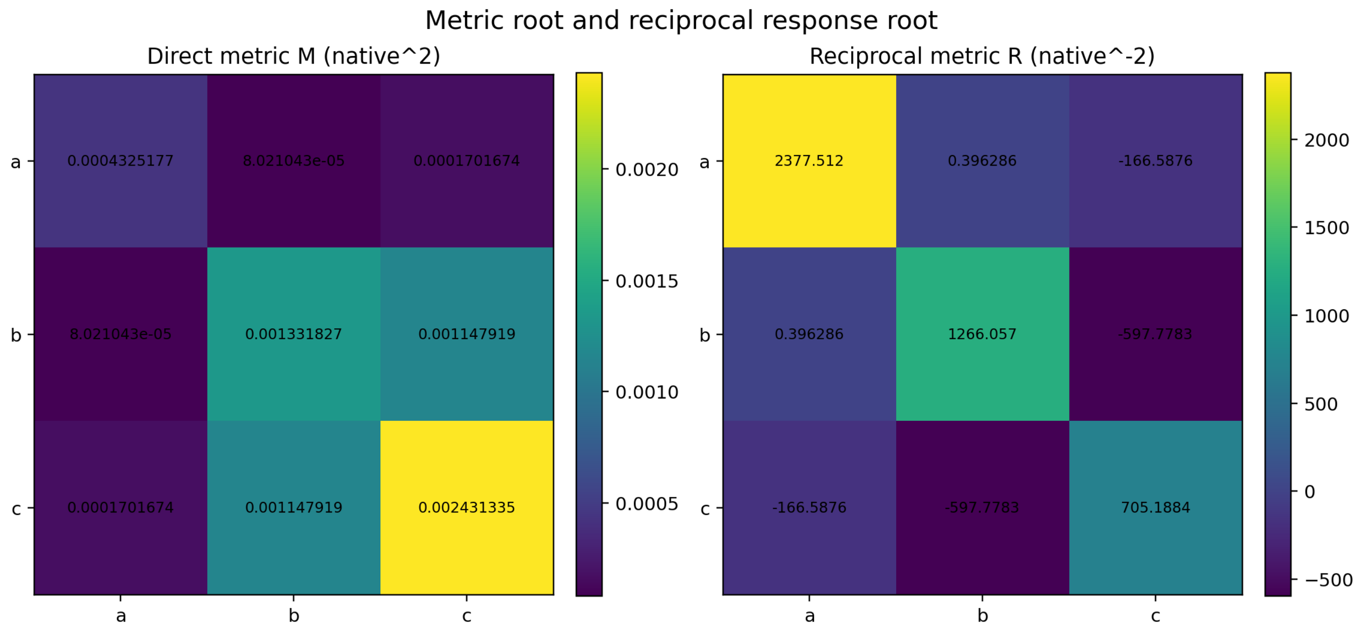

Figure 7. Direct and reciprocal metric matrices computed from the canonical Cell3 basis.

```
M = B^T B
R = M^-1
V_cell = sqrt(det(M))
d log(V_cell) = (1/2) trace(R dM)
```

The complete 351-pair squared-distance sum of the centered 27-node set is S2=121.5 trace(M)=0.509775038986547 native^2. Mixed coordinate sums cancel exactly, so this total is blind to off-diagonal shear. An edge-preserving rectilinear comparison has the same trace and the same S2, but Cell3 has volume ratio 0.759427120710 and reciprocal-trace gain 1.251733168861. The shear occupies 25.58 percent of the Frobenius-squared metric content while remaining invisible to the summed trace budget.

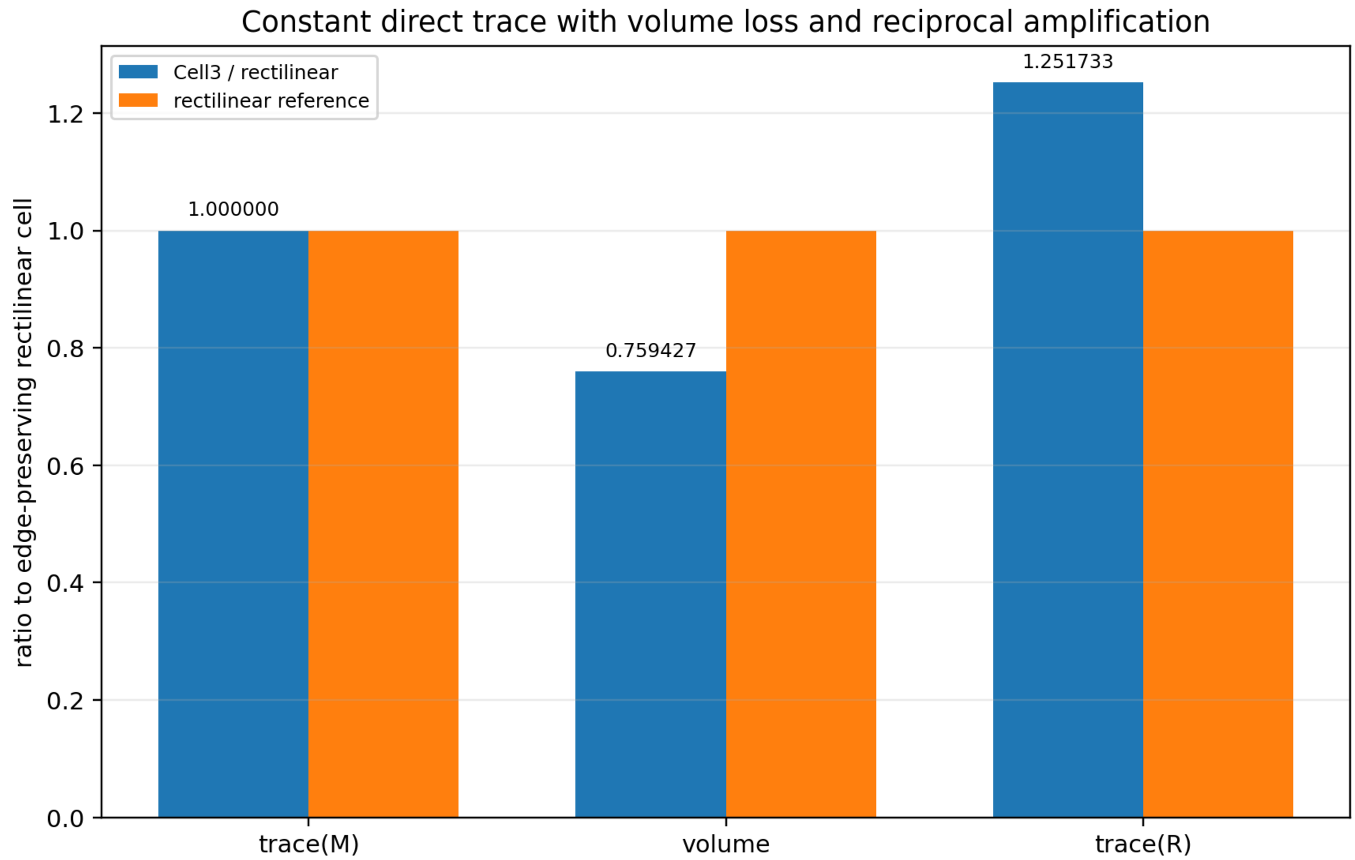

Figure 8. Dimensionless comparison with an edge-preserving rectilinear cell. Direct trace is conserved while volume and reciprocal response change.

4. The Anisotropic Carrier Tensor G_phase

The carrier transformation used by the calibrated force law is the fixed symmetric matrix G_phase. It converts the Earth-CMB translation vector V_E into g=G_phase V_E. Because G_phase is not proportional to the identity, it changes both magnitude and direction. Orientation is therefore a primary state variable rather than a secondary correction.

Table 6. G_phase matrix in ICRS coordinates.

Row	x	y	z
1	2.9764106378	1.2431507795	-2.0803600354
2	1.2431507795	2.5922286635	-2.3178839937
3	-2.0803600354	-2.3178839937	6.7735709133

The eigenvalues are all positive: 1.429328753, 2.112545026, and 8.800336435. The principal gain ratio is 6.156972:1. A component aligned with the highest-gain eigenvector receives more than six times the scalar tensor gain of a component aligned with the lowest-gain eigenvector. Positive definiteness means the quadratic form x^T G_phase x is positive for every nonzero x; signed force reversal still arises from spin orientation and the cross product.

Table 7. Principal gains and ICRS eigenvectors.

Mode	Eigenvalue	Unit eigenvector
1	1.42932875349	[-0.40895306728, 0.88452500943, 0.22443907068]
2	2.11254502645	[-0.82855389596, -0.25683230548, -0.497529505]
3	8.80033643467	[-0.38243408614, -0.38942608356, 0.83790900175]

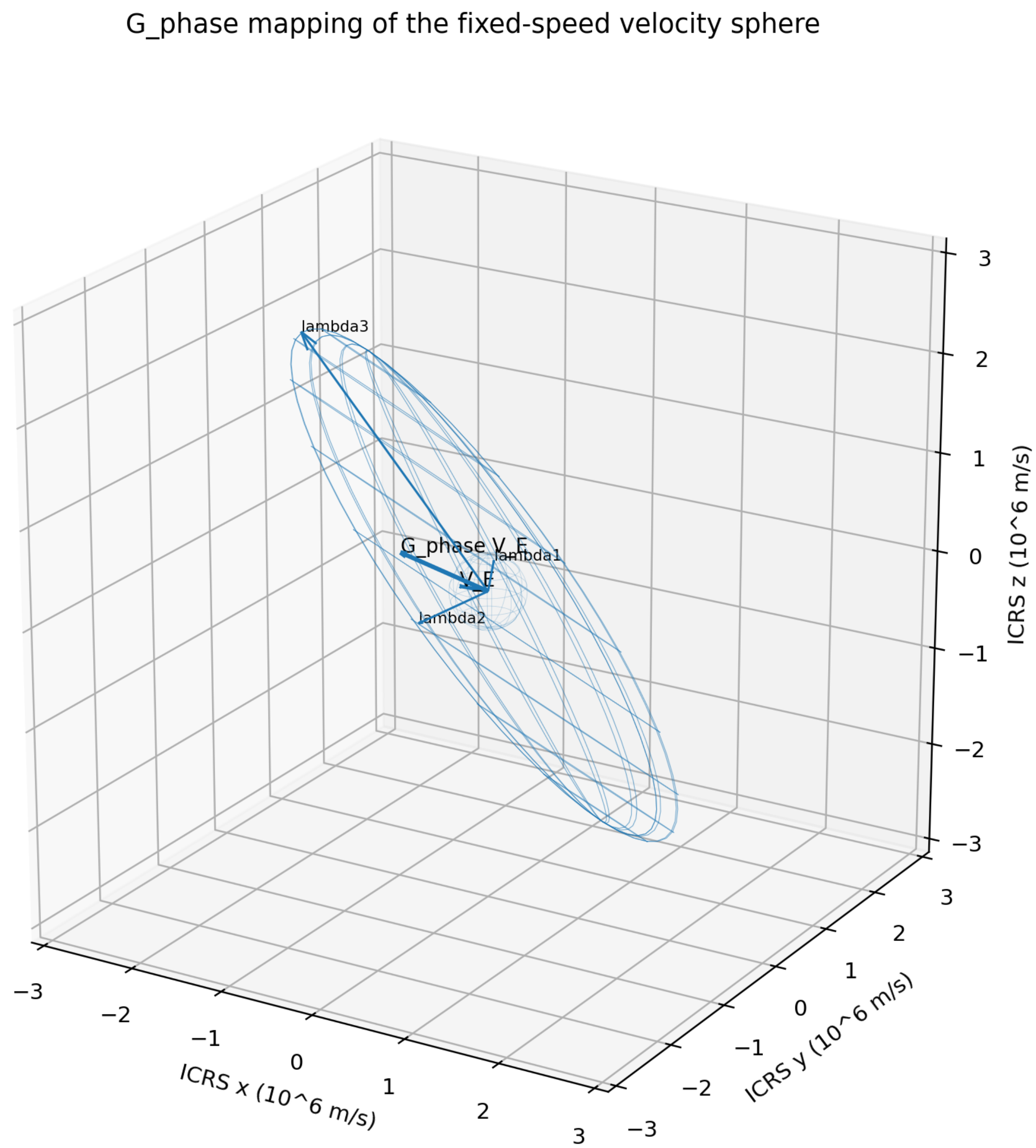

Figure 9. The fixed-speed velocity sphere and its G_phase image in ICRS velocity units.

4.1 Earth-CMB translation state

The declared Earth-CMB translation vector is V_E=[-3.590031990658e+05, 76679.886034, -44701.06010069] m/s with magnitude 369812.502000 m/s. The transformed vector is g=[-8.802219816351e+05, -1.439114365616e+05, 2.663350269486e+05] m/s with magnitude 930825.217270 m/s. The effective gain for this particular direction is |g|/|V_E|=2.517019333. This scalar is direction-specific; it is not a universal eigenvalue.

The angle between the Earth-CMB axis and g is 31.640861271 degrees. Their nonparallelism creates a component of the continuous-rim force parallel to the CMB axis even though the total force remains perpendicular to g.

Table 8. Translation-state transformation.

Object	Vector or definition	Magnitude/value
V_E	[-3.59003199066e+05, 76679.886034, -44701.0601007]	369812.502000 m/s
G_phase V_E	[-8.80221981635e+05, -1.43911436562e+05, 2.66335026949e+05]	930825.217270 m/s

Object	Vector or definition	Magnitude/value
Directional gain	|G_phase V_E|/|V_E|	2.517019333
Tensor condition	lambda_max/lambda_min	6.156971525

5. The Calibrated Constitutive Law

```
a_K = kappa_F H[omega cross (G_phase V_CMB)]
```

This compact equation separates carrier geometry from material response. G_phase and V_CMB define the transformed background state. omega defines the body spin state. H represents normalized mass geometry. kappa_F is the matter-interface susceptibility. None of these objects can be replaced by another without changing the model.

Table 9. Symbol and unit ledger for the constitutive law.

Symbol	Units/type	Role
kappa_F	dimensionless	material susceptibility at the interface
H	dimensionless tensor	body geometry and mass-distribution filter
omega	s^-1	directed angular velocity
G_phase	dimensionless 3x3 SPD tensor	anisotropic carrier transform
V_CMB	m/s	cosmologically referenced translation state
a_K	m/s^2	predicted whole-body acceleration

5.1 Weak matter coupling, not necessarily a weak carrier

The calibrated value is kappa_F=3.0740508997020e-11. Expressed as parts per trillion, the coupling is 30.7405090 ppt. Its inverse is 3.2530366e+10. The model therefore suppresses the material manifestation of the kinematic kernel by roughly 3.25e10. This result supports the description “strongly structured carrier, weak matter interface” more directly than “weak field.”

A small observable response can arise from a weak carrier, weak matter coupling, or both. The present calibration isolates kappa_F only because G_phase and the cosmic translation normalization are already declared. Without a dynamical field equation and an independently measured carrier amplitude, no intrinsic field stiffness follows from the coupling value.

5.2 Rank, nullspace, and maximum response

For a spherical body, the map from omega to acceleration is a scaled cross-product matrix. It has rank two and a one-dimensional nullspace spanned by g. Two independent spin components perpendicular to g produce response; the component parallel to g does not. This exact null orientation is a powerful experimental discriminator because it predicts a geometric zero rather than merely a small value.

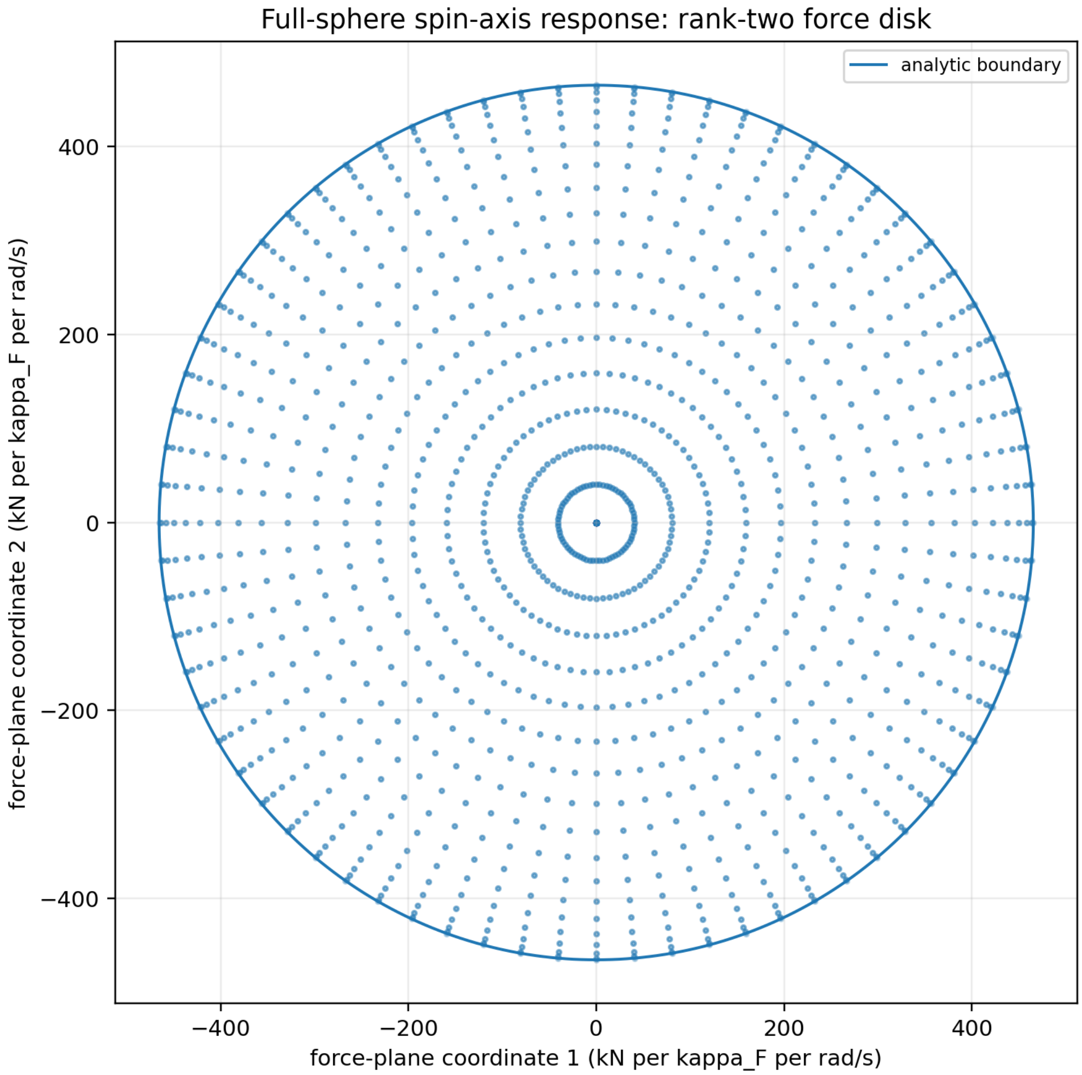

Figure 10. The full-sphere directed-axis map into the force plane. The center is the exact null and the circular boundary is the maximum response set.

5.3 Spin parity and time-odd behavior

Reversing the angular-velocity vector changes the sign of the leading response while preserving its magnitude envelope. Two otherwise identical bodies with opposite spin directions occupy opposite branches. An isotropic ensemble of randomly oriented spins has zero mean response vector even though individual members can have nonzero forces. The model is directionally selective but does not provide a universal one-way force without orientation control.

6. Body Geometry, Coherence, and Mass Scaling

The operator H prevents the constitutive law from treating all shapes as identical. A sphere has H=I/3. A continuous rim has an exact in-plane second moment. These examples show that the response depends on mass geometry, not only total mass. For geometrically similar bodies, force is extensive in mass while acceleration is independent of total mass because the mass factor cancels in F/M.

```
Sphere:       H = I/3
Uniform rim:  integral u u^T dm = (M/2)(I - n n^T)
```

The phrase coherent mass coupling means that mass elements add through an ordered tensor integral rather than through random independent phases. Coherence here is a property of the constitutive summation, not a claim of quantum coherence. The uniform rim is especially instructive because the full integral can be performed exactly.

6.1 Extensive force and mass-independent acceleration

```
F proportional to M
a = F/M  -> independent of M for equal normalized geometry H
```

This scaling can be tested by fabricating geometrically similar rotors with different total masses while holding angular rate and orientation fixed. A linear force-versus-mass relation and constant acceleration would match the leading-order model. Deviations would indicate composition dependence, nonlinear coupling, geometric corrections, or experimental systematics.

6.2 Shape as a second-moment filter

A second moment records how mass directions are distributed. In a sphere, directions are isotropic and the normalized tensor is I/3. In a thin rim, all mass lies in the plane perpendicular to n, so the tensor is proportional to the projector I-nn^T. The difference is measurable in principle: equal masses and equal spins can produce different responses if their normalized geometry tensors differ.

7. Exact Continuous-Rim Derivation

Consider a uniform zero-thickness circular rim of total mass M, radius R, and directed spin axis n. The source calculation constrains n to be perpendicular to the Earth-CMB direction for the ellipse study, but the analytic integral itself is valid for any directed unit n. Let u(theta) be the outward unit radial direction in the rim plane and t(theta)=n cross u(theta) the positive tangent.

```
r(theta) = R u(theta)
t(theta) = n cross u(theta)
v(theta) = V_E + R omega t(theta)
dm = (M/2pi) dtheta
```

7.1 Translation and rotational kernels

The anisotropic translation kernel B is linear in the tangent and transformed background velocity. The self kernel C is a positive quadratic reading of G_phase along the tangent. Under the declared radial-force rule, each mass element receives

```
g = G_phase V_E
B(theta) = t(theta)^T g                 [m/s]
C(theta) = t(theta)^T G_phase t(theta) [dimensionless]
dF = -kappa_F dm [omega B(theta) + R omega^2 C(theta)] u(theta)
```

Both bracketed terms have acceleration units. The first mixes spin with cosmic translation. The second depends only on rotational motion in the anisotropic metric. Their integrals behave differently under the antipodal map theta -> theta+pi.

7.2 Exact cancellation of the rotational self-term

At antipodal points, u and t reverse sign. Because C is quadratic in t, C does not change sign, while the radial vector u does. Therefore C u is odd across the rim and its complete integral is zero. A uniform rim does not generate a net R omega^2 self-resultant at this constitutive order.

```
integral_0^(2pi) C(theta) u(theta) dtheta = 0
```

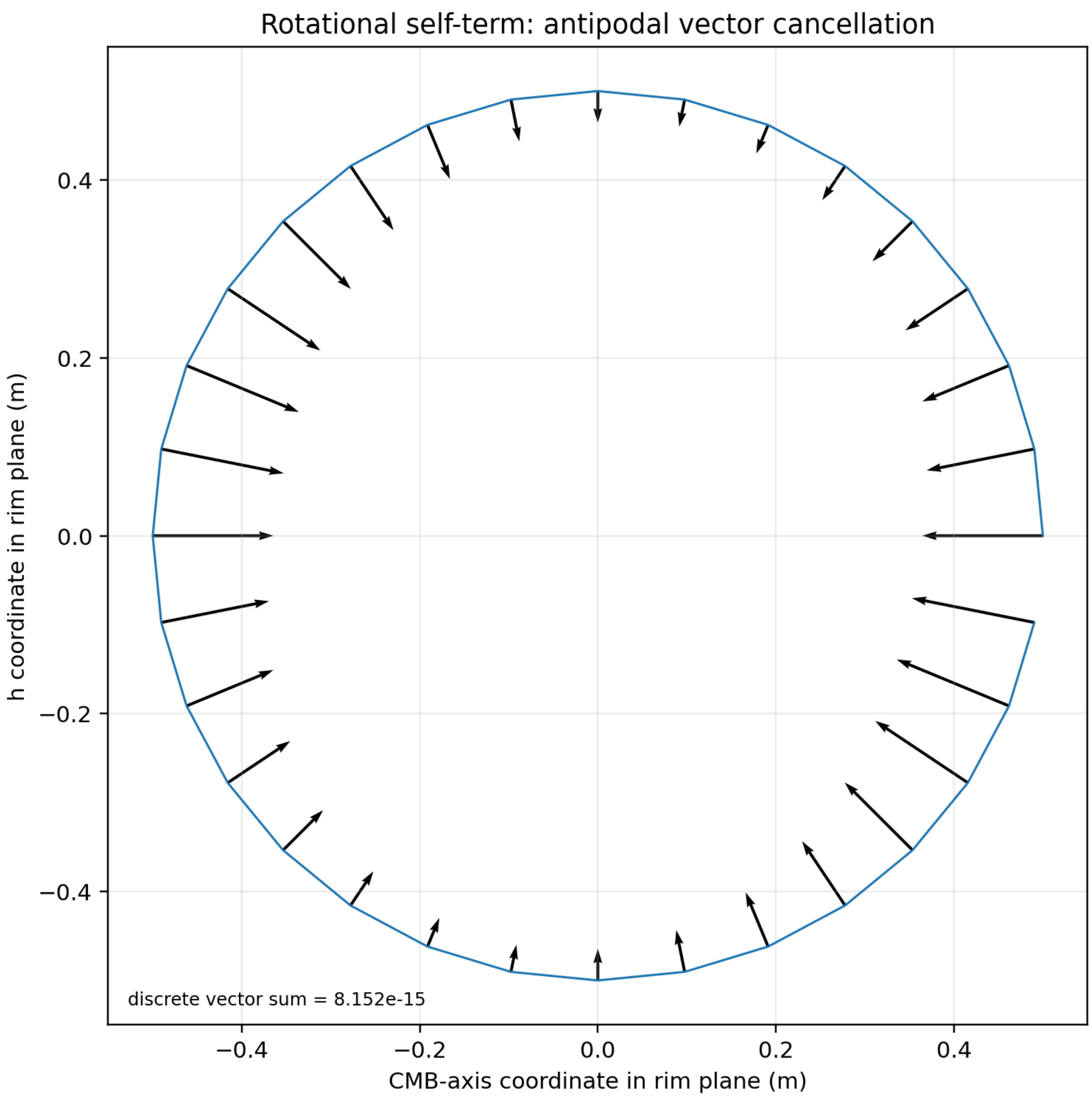

Figure 11. Actual self-term vector field at a maximum-force axis orientation. Antipodal radial vectors cancel.

7.3 Surviving translation-coupled integral

The translation term survives because B and u both reverse at the antipode, making their product even. The scalar triple-product identity gives B=u dot(g cross n). The rim second moment then closes the integral without a point-mass approximation.

```
B = (n cross u) dot g = u dot(g cross n)
integral u B dm = [integral u u^T dm](g cross n)
integral u u^T dm = (M/2)(I - n n^T)
(I - n n^T)(g cross n) = g cross n

F = (kappa_F M omega/2) n cross g
  = (kappa_F M omega/2) n cross(G_phase V_E)
```

The final force is independent of R at fixed omega. Radius appears in the canceled self-term but not in the surviving translation term. The force is perpendicular to g and to n. Because every elemental force is radial, r cross dF is zero locally, so the net torque about the rim center is also zero.

7.4 Force ellipse under the CMB-orthogonal axis constraint

When n is restricted to the great circle perpendicular to the CMB axis, the linear cross-product map sends that circle to an ellipse in the plane perpendicular to g. The exact source sweep uses 0.1-degree directed-axis spacing. Its semimajor and semiminor coefficients are listed below.

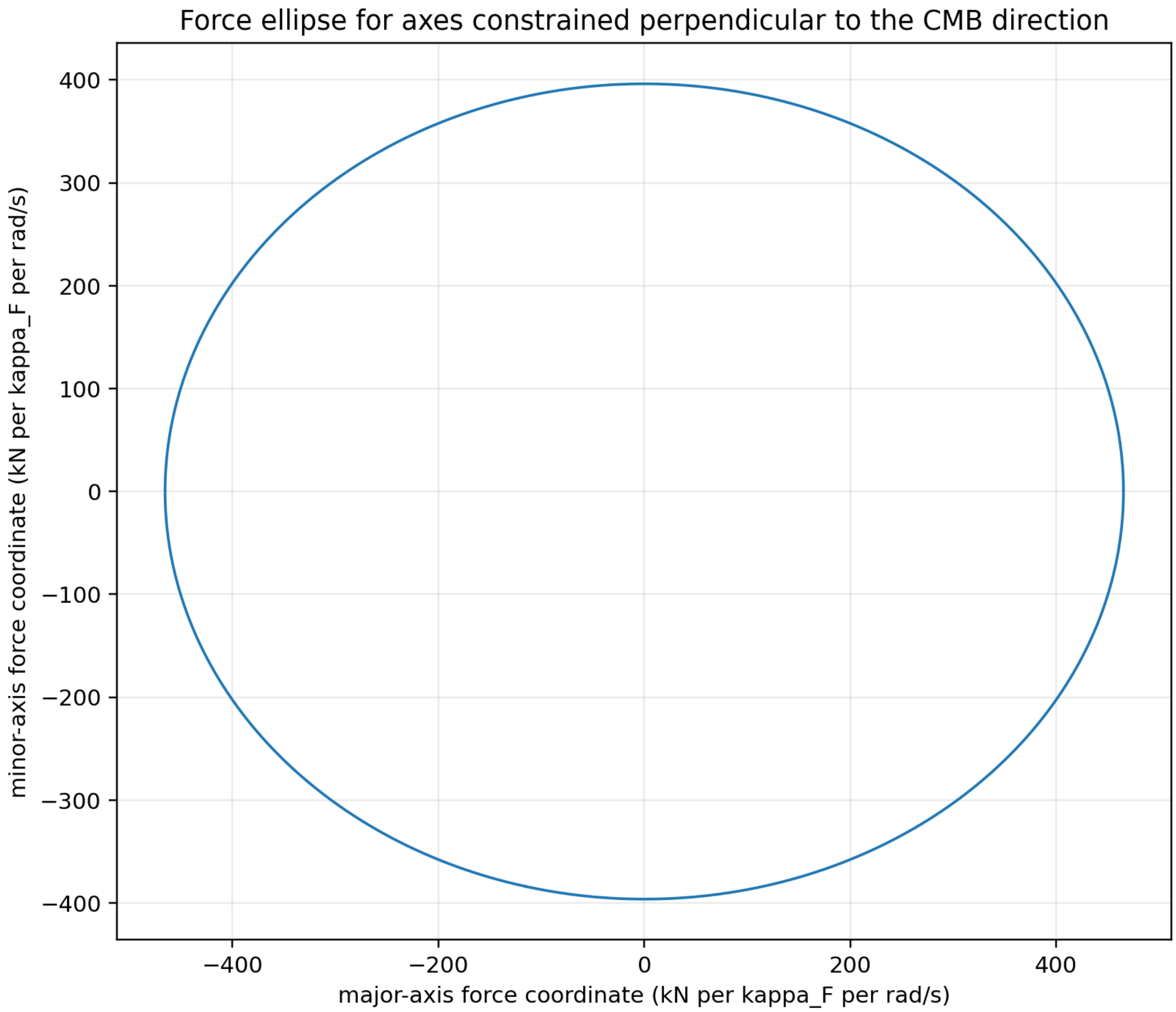

Figure 12. Exact constrained-axis force ellipse in its principal plane.

Table 10. Continuous-rim force-envelope invariants.

Quantity	Value	Units
Semi-major	4.65412608635217e+05	N per (kappa_F rad/s)
Semi-minor	3.96230434488576e+05	N per (kappa_F rad/s)
RMS magnitude	4.32207967008733e+05	N per (kappa_F rad/s)
CMB-parallel amplitude	2.44152286619709e+05	N per (kappa_F rad/s)
Full-cycle mean vector	[0, 0, 0]	N per (kappa_F rad/s)
Torque	0	N m

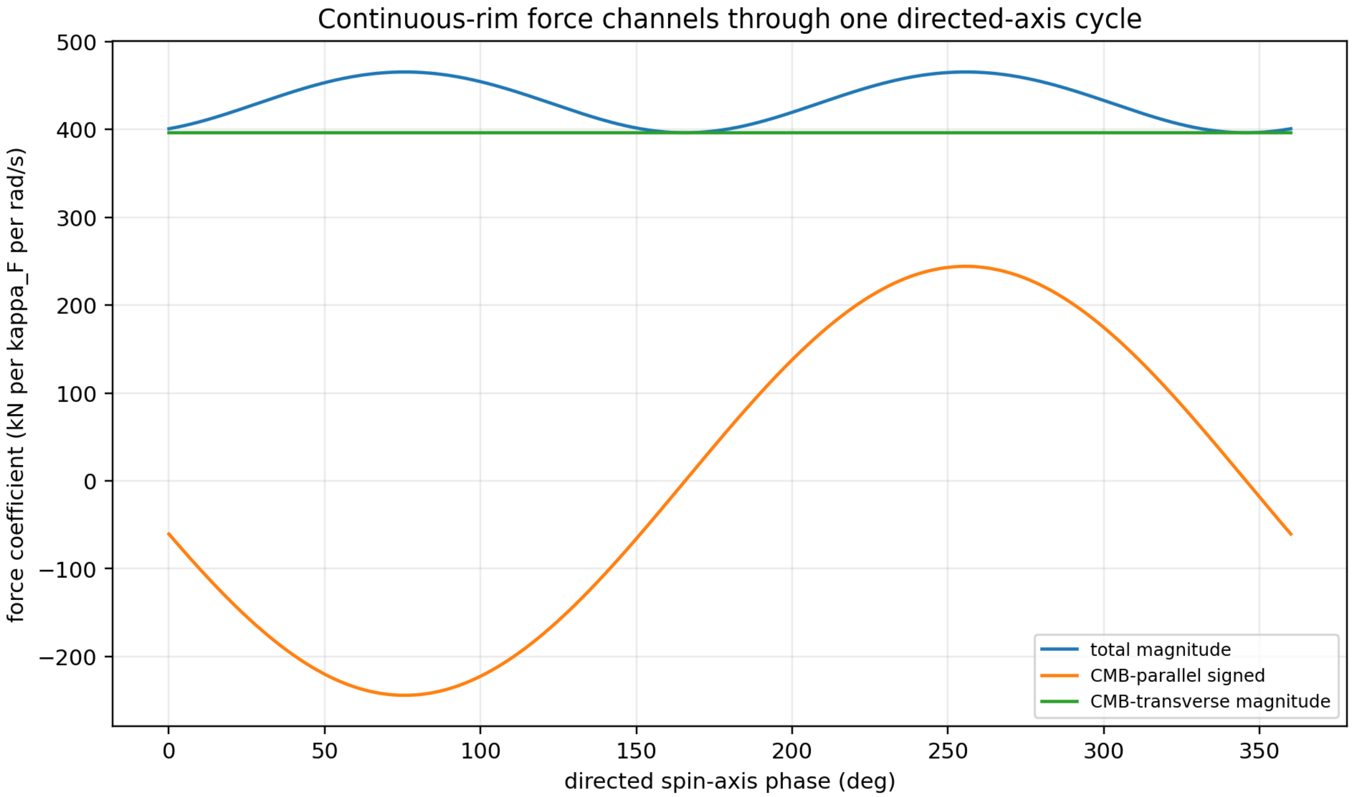

Figure 13. Total, CMB-parallel, and CMB-transverse channels through a complete directed-axis cycle.

8. Scaling Laws and Kinematic Impedance

The exact rim law separates three scaling variables: total mass, angular rate, and coupling. At fixed angular rate, the ideal rim force is independent of radius. Different engineering constraints reintroduce radius through the relation between angular rate and the constrained quantity.

Table 11. Continuous-rim scaling laws.

Constraint	Derived law	Interpretation
General	F proportional to kappa_F M omega	linear in coupling, mass, and angular rate
Fixed omega	F independent of R	leading ideal rim law is scale-free in radius
Fixed rim speed v	omega=v/R, so F proportional to 1/R	smaller radius requires higher angular rate
Fixed angular momentum L	omega=L/(M R^2), so F proportional to L/R^2	inverse-square radius scaling
Fixed rotational energy E	omega=sqrt(2E/(M R^2)), so F proportional to sqrt(M E)/R	inverse-radius scaling with energy constraint

8.1 Correct dimensional definition of interface impedance

For a one-kilogram rim at maximum orientation, the forward spin-to-force slope is dF/domega=kappa_F(465412.608635)=1.430702048308e-05 N per (rad/s). The inverse slope is domega/dF=6.989576e+04 (rad/s)/N. Keeping these reciprocal quantities separate prevents a unit inversion.

Table 12. Calibrated one-kilogram maximum-orientation engineering normalization.

Quantity	Value
Force per 1 rad/s	14.307020 microN
Force per 1 RPM	1.498228 microN
RPM for 1 N	667455.296 RPM
RPM for 9.80665 N	6545500.474 RPM
Inverse slope	6.989576e+04 (rad/s)/N

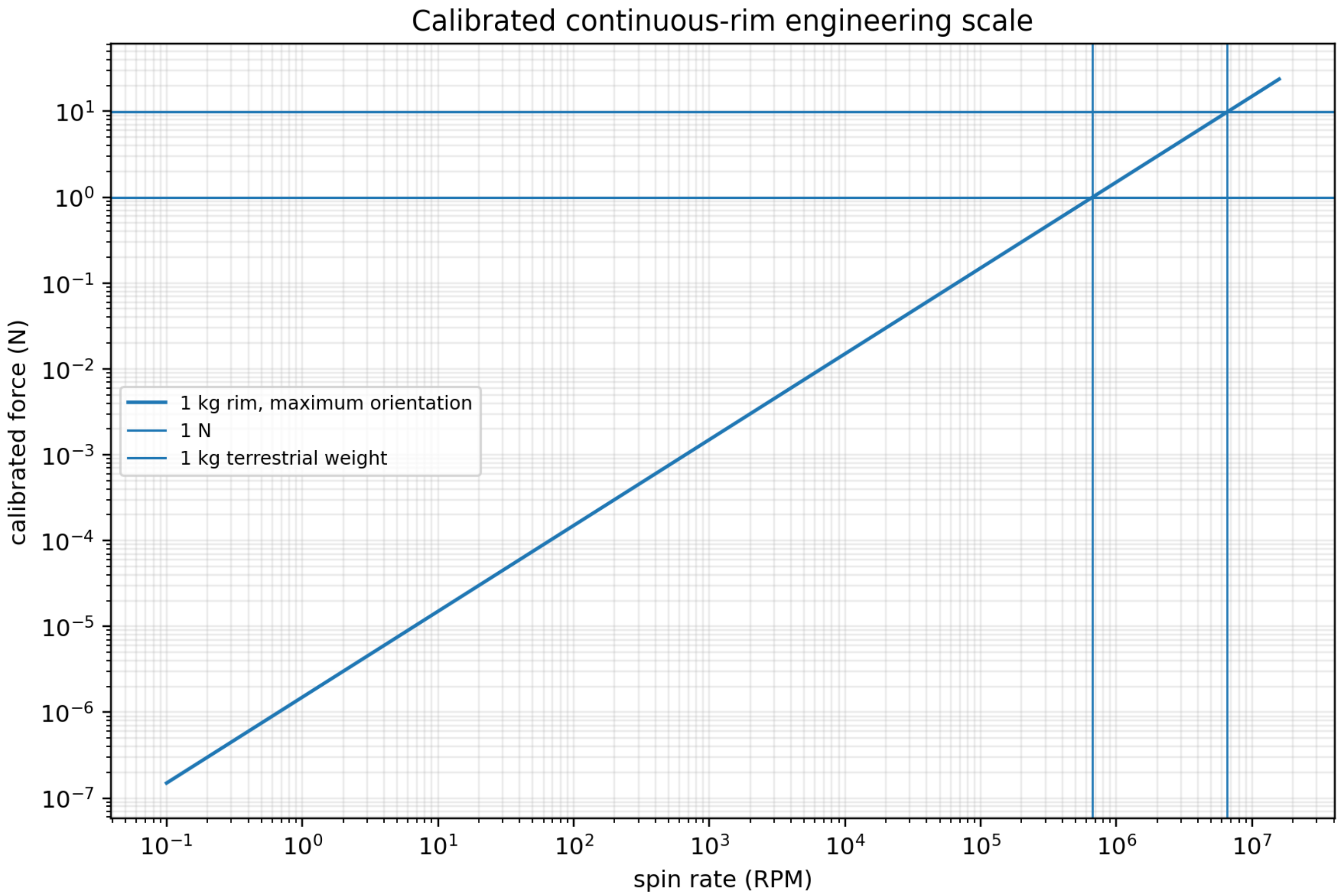

Figure 14. Calibrated force scale across spin rate for the reference one-kilogram rim.

The large angular rate required for ordinary newton-scale force motivates the term high kinematic impedance. This is an operational interface property: it is the inverse of a spin-to-force slope. It is not a Young modulus, bulk modulus, spring constant, or measure of resistance to field deformation.

8.2 Scale-free leading order and its limits

No wavelength, penetration depth, correlation length, or interaction range appears in the leading rim law. The result is therefore scale-free at the present constitutive order. This absence does not prove that the underlying field has no length scale. It shows only that the declared leading kinematic law contains none. Finite thickness, gradients across the body, higher-order derivatives, saturation, or a dynamical field equation could introduce additional scales.

9. Earth Kinematic Calibration of kappa_F

The Earth calibration uses two internally independent kinematic channels. The first projects the observed one-year ICRS eccentricity-vector change onto the computed unit-coupling K response. The second matches the observed fitted longitude-of-perihelion rate to the calculated unit-coupling apsidal response. The recommended coupling is the midpoint of the two values.

9.1 Full-vector projection

```
kappa_vector = (E_K dot e_dot_obs)/(E_K dot E_K)
```

The observed one-year eccentricity-vector change is [-8.205803672576e-07, -5.717914785385e-07, -2.880126483288e-07] yr^-1. The unit-coupling response is [-29838.8266503, -5080.14081201, -2201.13018941] yr^-1 per kappa_F. Their least-squares projection gives kappa_vector=3.0427399447546e-11. The fitted K-indexed vector is [-9.079178975348e-07, -1.545754737368e-07, -6.697466750922e-08] yr^-1, leaving residual [8.733753027714e-08, -4.172160048017e-07, -2.210379808195e-07] yr^-1.

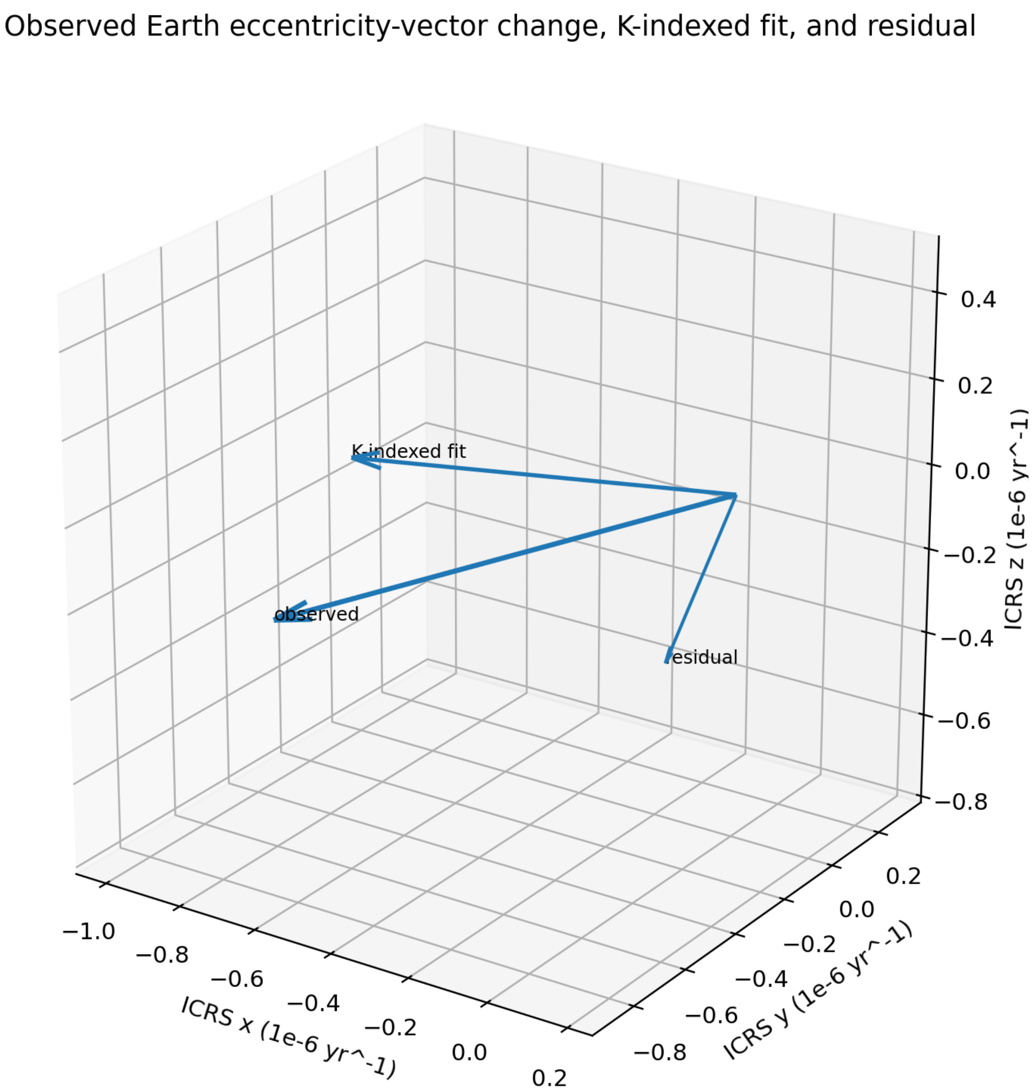

Figure 15. Observed, fitted, and residual Earth eccentricity-vector changes in ICRS.

The fitted and observed directions are separated by 27.473737 degrees. The K-indexed component contains 78.716 percent of the observed vector power. The residual shows that a single scalar coupling does not reproduce the complete observed vector under the Earth-only attribution.

9.2 Apsidal-rate cross-check

```
kappa_apsidal = observed varpi rate / (unit K varpi response)
```

The fitted observed apsidal rate is 5.6421894029068e-05 rad/yr. The calculated unit response is 1.8169185e+06 rad/yr per kappa_F, giving kappa_apsidal=3.1053618546494e-11.

9.3 Recommended coupling and interpretation of the interval

```
kappa_F = 0.5(3.0427399447546e-11 + 3.1053618546494e-11)
        = 3.0740508997020e-11
```

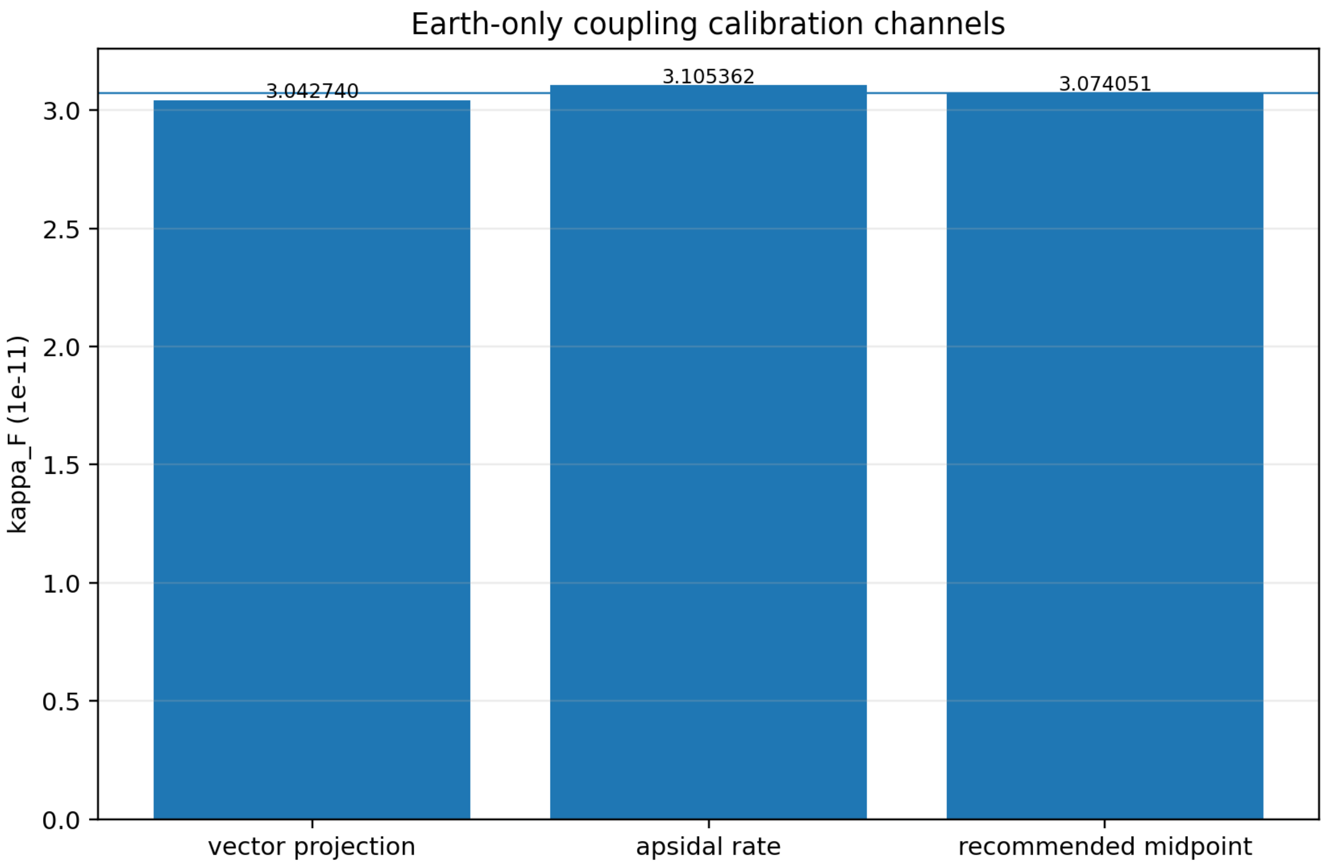

Figure 16. Earth-only vector and apsidal calibration channels with their midpoint.

The cross-channel fractional spread is 2.037 percent. This spread is an internal consistency measure between two model-attributed channels. It is not a statistical uncertainty and does not include ephemeris covariance, conventional-model uncertainty, unmodeled perturbations, or experimental systematics.

Table 13. Earth coupling calibration ledger.

Channel	kappa_F	Meaning
Vector projection	3.0427399447546e-11	full ICRS eccentricity-vector channel
Apsidal rate	3.1053618546494e-11	scalar longitude-of-perihelion channel
Recommended midpoint	3.0740508997020e-11	central Earth coupling
Interval	[3.04273994e-11, 3.10536185e-11]	cross-channel consistency range
Fractional spread	2.037114%	relative to channel mean

10. Annual Earth Response and Temporal Modulation

The 366-epoch Earth-Sun unit-response magnitude varies from 18.6591580328 to 23.2831655891 m/s^2 per unit coupling. The ratio is 1.247814373, corresponding to plus or minus 11.025 percent about the midpoint. kappa_F is held constant; the modulation is generated by changing orbital velocity and geometry relative to the fixed anisotropic carrier.

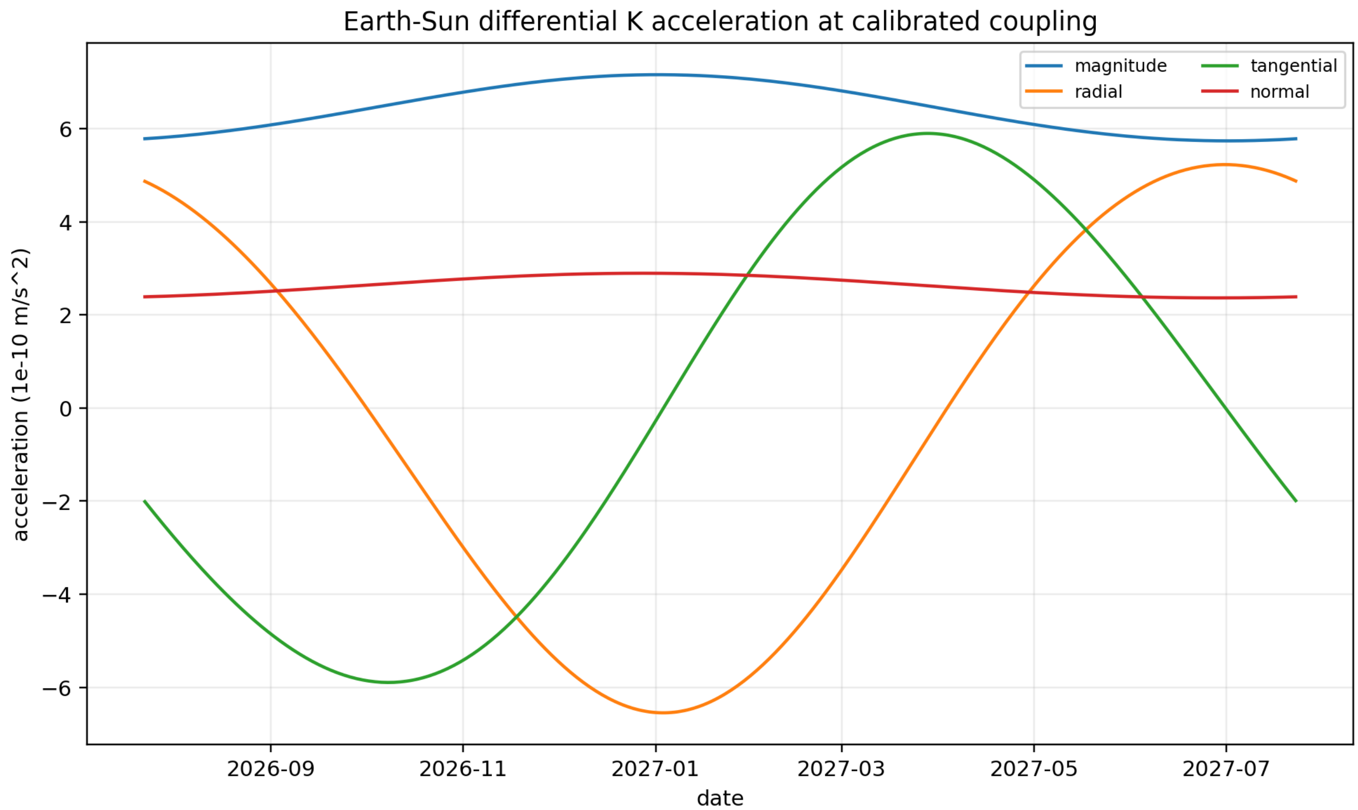

Figure 17. Calibrated daily Earth-Sun differential acceleration resolved into radial, tangential, normal, and magnitude channels.

Table 14. Calibrated Earth response scale.

Quantity	Value	Units
Minimum magnitude	5.7359201538411e-10	m/s^2
Maximum magnitude	7.1573636127084e-10	m/s^2
Maximum |tangential|	5.8956611826450e-10	m/s^2
Earth mass used	5.97219000000e+24	kg
Force range	3.42560049836e+15 to 4.27451353942e+15	N

10.1 Near-quadrature orbital response

The daily integration gives a force-to-semimajor-axis phase lag of 89.599664 degrees. A tangential acceleration changes orbital energy; the accumulated orbital-size response therefore peaks after the forcing peak. A near-90-degree relation is consistent with this integrated transverse geometry and should not be used alone as evidence for a spring-like energy-storage mechanism.

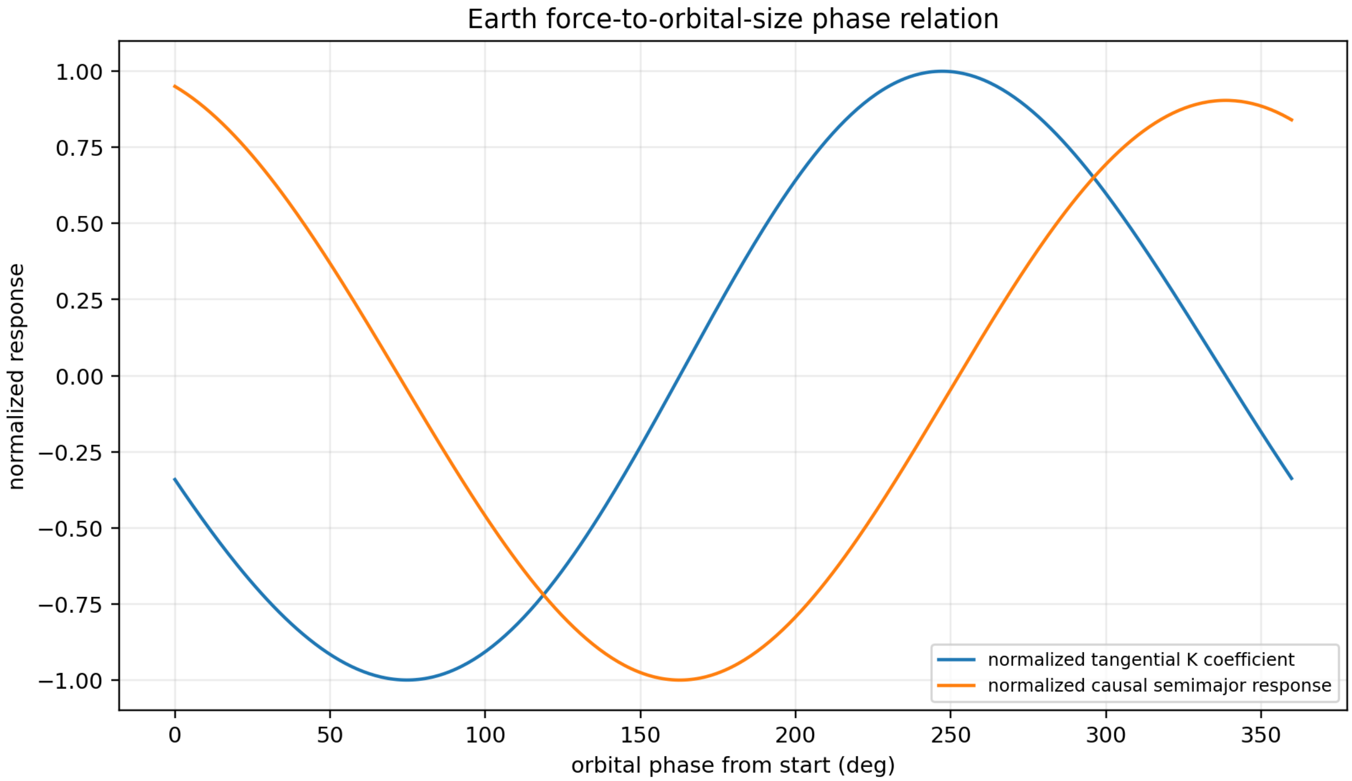

Figure 18. Normalized tangential K coefficient and normalized causal semimajor-axis response versus orbital phase.

10.2 Residual directional structure

The close scalar calibration values support a dominant approximately common gain across the two selected orbital observables. The remaining vector residual may represent conventional dynamics not included in the attribution, higher-order K-Field tensor terms, temporal or spatial phase structure, body-shape corrections, or limitations of assigning the full selected signal to Earth alone. The present data do not discriminate among these possibilities.

11. Consolidated Derived Properties

The following properties are consequences of the declared geometry, constitutive law, continuous-rim integral, and Earth calibration. Each statement is narrower than a complete field theory: it describes the present model at its calibrated leading order.

Table 15. Complete leading-order property map.

Property	Quantitative signature	Interpretation
Weak matter-interface coupling	kappa_F=3.0740508997e-11; inverse=3.25304e+10	Observable matter response is strongly suppressed relative to the unit kernel.
Strong anisotropy	eigenvalues=1.4293, 2.1125, 8.8003; ratio=6.157:1	One scalar carrier amplitude cannot represent the directional structure.
Positive-definite carrier transform	all G_phase eigenvalues are positive	The tensor quadratic form has no negative-gain principal direction.
Cosmological reference	explicit V_CMB dependence	The leading response is referenced to the declared CMB translation state.
Transverse/gyroscopic kernel	omega cross(G_phase V)	Response is perpendicular to spin and transformed translation before a non-isotropic H filter.
Exact null orientation	omega parallel G_phase V	Provides a geometric zero-response axis.
Spin-direction sensitivity	omega -> -omega gives a_K -> -a_K	Clockwise and counterclockwise states occupy opposite branches.
Mass extensivity	F proportional to M	Geometrically similar bodies have mass-independent acceleration.
Shape-tensor dependence	sphere H=I/3; rim second moment=(M/2)(I-nn^T)	Mass distribution is a constitutive state variable.
Zero rim torque	r cross dF=0 locally	The analyzed radial force law translates the rim without spin braking at this order.
No continuous-rim self-runaway	integral C(theta)u(theta)dtheta=0	The net uniform-rim force is linear in omega, not quadratic.
Scale-free leading rim law	no R at fixed omega	No intrinsic field length scale appears in this constitutive order.
Zero isotropic orientation mean	full directed-axis mean vector=0	Orientation selection is required for a nonzero ensemble mean.
Approximately constant scalar gain	Earth channels differ by 2.037%	Temporal modulation is geometric while kappa_F remains fixed.
High kinematic impedance	dF/domega=1.43070e-05 N/(rad/s)	Large spin rate is required for engineering-scale force at kilogram mass.

11.1 Paradigm implication within the model

The conceptual shift proposed by the framework is from a single scalar field strength to a layered description: exact discrete geometry, anisotropic carrier tensor, cosmological translation state, directed spin, normalized body geometry, and weak material susceptibility. In this description, a small observed force does not imply a weak background. It can be the product of a substantial structured kernel and a very small interface coefficient.

If independent controlled experiments reproduce the predicted null orientations, spin reversal, mass scaling, shape dependence, annual sidereal modulation, and material-independent kappa_F, the framework would open a new class of tensor-kinematic field models. If those discriminators fail, the model can be rejected or constrained without ambiguity because the leading symmetries are explicit.

12. What the Present Calibration Does Not Determine

The equations treat G_phase and V_CMB as prescribed background quantities. They do not change when a body interacts with them. This is equivalent to an effectively rigid background approximation, but it is not a measurement of rigidity. The same behavior can result from omitting the backreaction equation. A complete field theory requires additional state variables and dynamics.

Table 16. Undetermined field properties and required new information.

Undetermined property	Required information
Elastic modulus or static stiffness	field displacement variable and restoring-energy law
Energy density	Hamiltonian or stress-energy tensor
Propagation speed	time-dependent field equation
Interaction range	spatial Green function or gradient term
Dispersion	frequency-dependent response measurements and dynamic law
Hysteresis or memory	transient time-domain experiments
Saturation threshold	high-omega, high-velocity nonlinear measurements
Backreaction	reciprocal field momentum equation
Composition dependence	multiple-material tests at equal geometry
Screening or shielding	controlled intervening-media experiments
Thermal dependence	temperature-controlled coupling tests
Quantum response	microscopic coupling operator

A field Lagrangian L_K(phi, partial_t phi, grad phi), or an equivalent stress-energy and propagation law, would connect the kinematic carrier description to intrinsic stiffness, energy, momentum, range, and wave behavior. Until such a law is supplied and experimentally constrained, these quantities remain outside the present result.

13. Experimental and Discovery Pathways

The most informative experiments are those that target exact symmetries rather than raw force magnitude. A successful program should first reproduce the predicted sign, null, phase, mass, and geometry relations before attempting a broad search for materials or devices.

Table 17. Discriminating experimental program.

Test	Controlled change	Leading prediction
Spin reversal	same body and orientation; reverse omega	force vector reverses; magnitude envelope preserved
Null-axis scan	align omega through G_phase V direction	single geometric zero at parallel and antiparallel axes
Perpendicular maximum	rotate omega into plane normal to G_phase V	maximum magnitude at 90 degrees
Mass scaling	geometrically similar bodies with varied M	F linear in M; a approximately constant
Geometry scaling	equal M and omega; sphere, disk, rim, compound shapes	response follows H rather than mass alone
Radius test	uniform rims, fixed omega, varied R	leading resultant independent of R
Fixed rim-speed test	uniform rims, fixed v_rim, varied R	F proportional to 1/R
Material comparison	equal geometry and mass with different composition	tests universality or composition dependence of kappa_F
Temperature sweep	fixed mechanical state, controlled temperature	tests thermal dependence
Shielding test	intervening media and enclosure changes	tests screening or environmental coupling
Frequency sweep	broad omega and modulation frequencies	tests dispersion, memory, and nonlinear saturation
Annual/sidereal scheduling	repeat fixed apparatus orientations over time	tests cosmic-frame modulation predicted by V and G_phase
Reaction measurement	simultaneous body force and support/environment momentum accounting	tests backreaction channel and conservation closure

13.1 Measurement hierarchy

A robust sequence is: establish instrument reversal symmetry; measure zero-spin baseline; execute spin-reversal pairs; scan through the predicted null; repeat with dummy masses and locked bearings; scale mass and geometry; compare materials; then search for temporal modulation. Force, torque, vibration, temperature, magnetic field, pressure, bearing state, drive current, and support reactions should be recorded synchronously.

The leading rim prediction is micro-newton scale at one radian per second for one kilogram. This magnitude is accessible to precision force metrology but is vulnerable to ordinary electromagnetic, thermal, aerodynamic, bearing, and vibration artifacts. The exact sign and null relations are therefore more valuable than a single positive reading.

14. Reproducible Computational Reconstruction

The complete leading-order reconstruction requires only the source basis, G_phase, the translation vector, body state, and coupling. The following sequence reproduces every central numerical result in this paper.

Table 18. Student reconstruction sequence.

Step	Action
1	Load the canonical Cell3 basis B and verify M=B^T B, R=M^-1, volume=|det(B)|.
2	Generate all L in {-1,0,1}^3 and compute p=(1/2)B L. Confirm 27 nodes and class counts 8/12/6/1.
3	Rotate the common basis into ICRS with the locked orthogonal rotation; regenerate nodes, vertices, faces, and planes.
4	Load symmetric G_phase and verify all three eigenvalues are positive.
5	Load V_E and compute g=G_phase V_E, its magnitude, direction, and gain relative to |V_E|.
6	For a body, evaluate the kernel omega cross g and apply its geometry operator H.
7	Multiply by kappa_F to obtain acceleration; multiply by mass for force.
8	For a uniform rim, integrate the elemental force or use F=(kappa_F M omega/2)n cross g.
9	Sweep directed n to reconstruct the force disk or constrained ellipse and verify spin reversal and zero mean.
10	For Earth, form the 366-epoch differential response, integrate the eccentricity-vector template, and calculate the vector and apsidal coupling channels.

```
Pseudo-code
-----------
B, G, V, kappa = load_source_arrays()
M = transpose(B) @ B
R = inverse(M)
nodes = [0.5 * B @ L for L in product([-1,0,1], repeat=3)]
g = G @ V
a = kappa * H @ cross(omega, g)
F_rim = 0.5 * kappa * M_rim * omega_scalar * cross(n, g)
```

14.1 Numerical precision and coordinate discipline

All vector operations must remain in one declared coordinate frame. A rigid rotation may be applied to all compatible vectors and tensors together, but mixing the canonical local basis with ICRS vectors without transformation is invalid. Symmetry, unit norms, handedness, determinant, eigenvalues, and dimensional units should be checked before computing forces.

15. Conclusions

The calibrated K-Field model is best characterized as a weakly matter-coupled, strongly anisotropic, positive-definite, cosmologically referenced, spin-direction-sensitive, transverse, mass-extensive, shape-dependent, and leading-order scale-free kinematic interaction. Its exact Cell3 foundation supplies the geometry; G_phase supplies directional carrier structure; V_CMB supplies the preferred translation state; omega supplies directed rotation; H supplies body geometry; and kappa_F supplies the small interface susceptibility.

The central numerical value is kappa_F=3.0740508997e-11. It produces an Earth-Sun differential acceleration range of 5.735920e-10 to 7.157364e-10 m/s^2 under the Earth attribution. For a one-kilogram continuous rim, the calibrated maximum-orientation force is 14.307020 micro-newtons per radian per second.

The strongest stiffness-related statement is operational: the matter-K-Field interface has high kinematic impedance. The calibration does not establish intrinsic elastic stiffness, energy density, resistance to deformation, propagation speed, range, or backreaction. Those properties require a dynamical field and energy law beyond the calibrated kinematic coupling.

Foundational result: the observable is controlled by structured carrier geometry multiplied by weak material susceptibility. The coupling coefficient and the carrier strength are distinct physical concepts.

Appendix A. Complete Cell3 Basis, Metric, and Dimension Ledger

Table 19. Canonical and ICRS basis vectors.

Frame	Vector	x	y	z	length
canonical local	a	0.020797058781610488	0	0	0.020797058781610488
canonical local	b	0.0038568161048841863	0.036289832979038449	0	0.036494205130587522
canonical local	c	0.0081822843518354026	0.03076238244505114	0.037657147000716373	0.049308565900292041
ICRS	a	0.020381438516946449	-0.0034164144964105494	-0.0023329659167289608	0.020797058782432949
ICRS	b	0.0021987459683233553	0.012902170258692541	-0.034066501536055713	0.036494205130727188
ICRS	c	-0.00063037948534737202	-0.024257722017127867	-0.042924354571093816	0.049308565899859422

Table 20. Direct and reciprocal metric entries.

Direct entry	Value (native^2)	Reciprocal entry	Value (native^-2)
M_aa	0.00043251765396576191	R_aa	2377.5121840862698
M_ab	8.0210431243138427e-05	R_ab	0.39628598570983992
M_ac	0.00017016744863297255	R_ac	-166.58755028383641
M_ba	8.0210431243138427e-05	R_ba	0.39628598570981771
M_bb	0.0013318270081134006	R_bb	1266.057440195837
M_bc	0.001147919287031111	R_bc	-597.77833398050063
M_ca	0.00017016744863297255	R_ca	-166.58755028383638
M_cb	0.001147919287031111	R_cb	-597.77833398050063
M_cc	0.0024313346711434432	R_cc	705.18842087050734

Table 21. Metric invariants and edge-preserving rectilinear comparison.

Quantity	Value
trace(M)	0.0041956793332226057
S2=121.5 trace(M)	0.50977503898654664
det(M)	8.0773444784317332e-10
volume	2.8420669377113084e-05
trace(R)	4348.7580451526137
rectilinear volume	3.7423827253530628e-05
Cell3/rect volume	0.75942712071042462
rectilinear trace(R)	3474.1893506825418
Cell3/rect trace(R)	1.2517331688608895
off-diagonal Frobenius fraction	0.25582394844780321

Appendix B. Complete 27-Node Registry

The node coordinate is p=(1/2)B_ICRS L. Lambda is the integer-index sum i+j+k. Polarity follows chi=(-1)^(i+j+k).

Table 22. All centered Cell3 nodes in ICRS native coordinates.

L	Class	lambda	chi	x	y	z
-1,-1,-1	corner	-3	-	-0.010974902499961	0.0073859831274229	0.039661911011939
-1,-1,0	edge center	-2	+	-0.011290092242635	-0.004742877881141	0.018199733726392
-1,-1,1	corner	-1	-	-0.011605281985309	-0.016871738889705	-0.0032624435591546
-1,0,-1	edge center	-2	+	-0.0098755295157995	0.013837068256769	0.022628660243911
-1,0,0	face center	-1	-	-0.010190719258473	0.0017082072482053	0.0011664829583645
-1,0,1	edge center	0	+	-0.010505909001147	-0.010420653760359	-0.020295694327182
-1,1,-1	corner	-1	-	-0.0087761565316379	0.020288153386115	0.0055954094758835
-1,1,0	edge center	0	+	-0.0090913462743115	0.0081592923775515	-0.015866767809663
-1,1,1	corner	1	-	-0.0094065360169852	-0.0039695686310124	-0.03732894509521
0,-1,-1	edge center	-2	+	-0.00078418324148799	0.0056777758792177	0.038495428053575
0,-1,0	face center	-1	-	-0.0010993729841617	-0.0064510851293463	0.017033250768028
0,-1,1	edge center	0	+	-0.0014145627268354	-0.01857994613791	-0.0044289265175191
0,0,-1	face center	-1	-	0.00031518974267369	0.012128861008564	0.021462177285547
0,0,0	body center	0	+	0	0	0
0,0,1	face center	1	-	-0.00031518974267369	-0.012128861008564	-0.021462177285547
0,1,-1	edge center	0	+	0.0014145627268354	0.01857994613791	0.0044289265175191
0,1,0	face center	1	-	0.0010993729841617	0.0064510851293463	-0.017033250768028
0,1,1	edge center	2	+	0.00078418324148799	-0.0056777758792177	-0.038495428053575
1,-1,-1	corner	-1	-	0.0094065360169852	0.0039695686310124	0.03732894509521
1,-1,0	edge center	0	+	0.0090913462743115	-0.0081592923775515	0.015866767809663
1,-1,1	corner	1	-	0.0087761565316379	-0.020288153386115	-0.0055954094758835
1,0,-1	edge center	0	+	0.010505909001147	0.010420653760359	0.020295694327182
1,0,0	face center	1	-	0.010190719258473	-0.0017082072482053	-0.0011664829583645
1,0,1	edge center	2	+	0.0098755295157995	-0.013837068256769	-0.022628660243911
1,1,-1	corner	1	-	0.011605281985309	0.016871738889705	0.0032624435591546
1,1,0	edge center	2	+	0.011290092242635	0.004742877881141	-0.018199733726392
1,1,1	corner	3	-	0.010974902499961	-0.0073859831274229	-0.039661911011939

Appendix C. Vertices, Faces, and Plane Equations

Table 23. Eight centered boundary vertices.

Primitive coefficients	ICRS x	ICRS y	ICRS z
[-0.5, -0.5, -0.5]	-0.0109749024999612	0.00738598312742294	0.0396619110119392
[-0.5, -0.5, 0.5]	-0.0116052819853086	-0.0168717388897049	-0.00326244355915457
[-0.5, 0.5, -0.5]	-0.00877615653163786	0.0202881533861155	0.00559540947588353
[-0.5, 0.5, 0.5]	-0.00940653601698523	-0.00396956863101239	-0.0373289450952103
[0.5, -0.5, -0.5]	0.00940653601698523	0.00396956863101239	0.0373289450952103
[0.5, -0.5, 0.5]	0.00877615653163786	-0.0202881533861155	-0.00559540947588353
[0.5, 0.5, -0.5]	0.0116052819853086	0.0168717388897049	0.00326244355915457
[0.5, 0.5, 0.5]	0.0109749024999612	-0.00738598312742294	-0.0396619110119392

Table 24. Face areas, included angles, and ICRS unit normals.

Face	Area (native^2)	Angle	Unit normal
A_ab	0.000754721789672624	83.933483012313 deg	[0.194092200808, 0.913177168821, 0.358379234791]
A_ac	0.001011255757688065	80.448130018338 deg	[0.0890525941016, 0.866576769291, -0.491033948326]
A_bc	0.001385784431733287	50.363231813756 deg	[-0.995965190309, 0.0836021625074, -0.0326192905644]

Table 25. All six boundary plane equations.

Plane family	Side	Unit normal n	d (native)	Equation
ab faces (constant c coefficient)	1	[0.1940922008, 0.9131771688, 0.3583792348]	-0.01882857350019	n dot x + d = 0
ab faces (constant c coefficient)	2	[0.1940922008, 0.9131771688, 0.3583792348]	0.01882857350019	n dot x + d = 0
ac faces (constant b coefficient)	1	[0.0890525941, 0.8665767693, -0.4910339483]	0.01405216690339	n dot x + d = 0
ac faces (constant b coefficient)	2	[0.0890525941, 0.8665767693, -0.4910339483]	-0.01405216690339	n dot x + d = 0
bc faces (constant a coefficient)	1	[-0.9959651903, 0.08360216251, -0.03261929056]	-0.01025436161905	n dot x + d = 0
bc faces (constant a coefficient)	2	[-0.9959651903, 0.08360216251, -0.03261929056]	0.01025436161905	n dot x + d = 0

Appendix D. Complete Formula Ledger

Table 26. Canonical formulas used in this reference.

Object	Formula
Node embedding	p=(1/2)B L; L in {-1,0,1}^3
Direct metric	M=B^T B
Reciprocal metric	R=M^-1
Cell volume	V_cell=sqrt(det(M))
Pair squared distance	d_ij^2=(1/4)(Delta L)^T M(Delta L)
Squared-distance budget	S2=121.5 trace(M)
Volume derivative	d log(V_cell)=(1/2)trace(R dM)
Carrier vector	g=G_phase V_CMB
Whole-body law	a_K=kappa_F H[omega cross g]
Sphere geometry	H=I/3
Rim mass measure	dm=(M/2pi)dtheta
Rim second moment	integral u u^T dm=(M/2)(I-nn^T)
Rim self cancellation	integral C(theta)u(theta)dtheta=0
Continuous-rim force	F=(kappa_F M omega/2)n cross g
Rim torque	tau=integral r cross dF=0
Vector calibration	kappa=(E_K dot e_dot_obs)/(E_K dot E_K)
Apsidal calibration	kappa=varpi_dot_obs/(varpi_dot_K per unit kappa)
Recommended coupling	mean of vector and apsidal channels

Appendix E. Representative Earth Daily Ledger

The machine-readable companion contains all 366 daily records. This table provides approximately monthly checkpoints plus the closing epoch.

Table 27. Representative Earth-Sun response records.

Date	Phase (deg)	q_R/unit	q_T/unit	q_N/unit	|q|/unit	calibrated |a|
2026-07-23	0.000000	15.831520001	-6.5442976789	7.7642698711	18.808209492	5.7817393309e-10
2026-08-22	28.749946	10.996785318	-13.907814749	8.0296907158	19.463620717	5.9832160777e-10
2026-09-21	57.844907	3.1625455128	-18.410433586	8.4287180826	20.493634316	6.2998475007e-10
2026-10-22	88.409632	-6.614898849	-18.65707429	8.8790733306	21.695189584	6.6692117059e-10
2026-11-21	118.483810	-15.398291148	-13.871442946	9.2337342666	22.688899228	6.9746831085e-10
2026-12-22	149.938457	-20.775316176	-4.4835068871	9.4069475374	23.242337623	7.1448128882e-10
2027-01-21	180.502815	-20.235080544	6.2633225758	9.334176995	23.147668446	7.1157111012e-10
2027-02-21	211.920564	-13.715108502	15.260103059	9.0331385677	22.418129694	6.8914471755e-10
2027-03-24	242.926922	-3.5982888453	19.12061271	8.5968329523	21.27089678	6.5387819385e-10
2027-04-23	272.430869	6.2605645318	17.252673526	8.1674501016	20.088719551	6.1753746411e-10
2027-05-24	302.432117	13.695989631	10.814894645	7.8333380948	19.128598063	5.8802284087e-10
2027-06-23	331.143598	16.86928266	2.2991281744	7.6929132364	18.682601582	5.7431268203e-10
2027-07-23	359.755355	15.858026233	-6.4736249811	7.7628438208	18.80549283	5.7809042154e-10

Appendix F. Source and Reference Ledger

Table 28. Materialized proprietary source artifacts used for this report.

File	Bytes	SHA-256	Role
K-Field_mass_harmonics_1784575932.json	28761	a8a90ad5ad3c049c660af65eb245414b730f87d3217db12ae2bc24472a370210	machine-readable numerical source
K-Field_earth_kinematic_derivation_of_coupling_constant_1784828045.json	978067	2f9d605451b26f29a84ff7511dd69efa3d53db6212a586b0b6562dc54faf1477	machine-readable numerical source
K-Field_flywheel_force_analysis_1784808116.json	2316139	a965becf29845009e05ca992befb7699a5869415b5296c315bdf81fef10fa4e8	machine-readable numerical source
K-Field_flywheel_force_distribution_disk_1784810003.json	31792574	6501035abae5ed5ff29aaa60906be4dc03ac639d0834abb74a3474e2dee06156	machine-readable numerical source

Table 29. Public astronomical references identified by the source analysis.

ID	Reference	Location	Use
[R1]	NASA/JPL Solar System Dynamics, Approximate Positions of the Planets	https://ssd.jpl.nasa.gov/planets/approx_pos.html	Fitted Keplerian elements and rates; 1800-2050 interval.
[R2]	R. S. Park et al., The JPL Planetary and Lunar Ephemerides DE440 and DE441, Astronomical Journal 161:105 (2021)	https://ssd.jpl.nasa.gov/doc/de440_de441.html	Integrated ephemeris and observational-fit context.
[R3]	NASA/JPL Solar System Dynamics, Astrodynamic Parameters	https://ssd.jpl.nasa.gov/astro_par.html	DE440 mass-parameter context.
[R4]	NASA/JPL NAIF generic planetary constants kernel pck00011.tpc	https://naif.jpl.nasa.gov/pub/naif/generic_kernels/pck/pck00011.tpc	Planetary orientation and rotation constants used by the parent analysis.

Machine-readable companion: K-Field_derived_properties_1784893730.json. The companion contains all numerical arrays required to reproduce the report figures and central calculations, including the 27-node registry, six boundary planes, 0.1-degree rim sweep, and 366-epoch Earth ledger.

Archival colophon: generated at Unix epoch 1784893730 (2026-07-24T11:48:50Z) using Python ReportLab. Document fonts are restricted to Helvetica, Helvetica-Bold, Times-Roman, and Courier.